Project	**Mansourah Massarah Gold Project**
Customer:	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **DESIGN CALCULATION**
3320 LOW GRADE THICKENING
DESIGN OF FOUNDATION

Document Number:	**MD-216-3320-EG-CV-CAL-1210**
Revision:	**D01**
Revision Date:	**15.11.2023**
Status:	**AS BUILT**

Outotec Document Number:	900707	GAU01	ST02	CED01	33201	OU100665184	3
L&T Document Number:	O19045	450	C	K	24088	001	3

REVISION HISTORY

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
A01	29.01.2020	FIRST ISSUE	AVG	SJM	AG	
A02	24.03.2020	REVISED AS PER EMPLOYER COMMENTS DATED 16.03.2020	AVG	SJM	AG	
C01	22.04.2020	ISSUED FOR CONSTRUCTION	AVG	SJM	AG	
D01	15.11.2023	AS BUILT	AVG	SJM	AG	

MA'ADEN معادن Saudi Arabian Mining Company شركة التعدين العربية السعودية		Outotec LARSEN & TOUBRO
MD-216-3320-EG-CV-CAL-1210	**Mansourah Massarah Gold Project**	
900707_GAU01_ST02_CED01_ 33201_OU100665184_R3	**DESIGN CALCULATION LOW GRADE THICKENING DESIGN OF FOUNDATION**	**Date: 15.11.2023**
O19045_450_C_K_24088_001_R3	**AS BUILT**	**Rev: D01**

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DEFINITIONS AND ABBREVIATIONS

Employer	MA'ADEN GOLD AND BASE METALS or their representatives
EPC Contractor	Consortium of Outotec (OT) and Larsen & Toubro (L&T)
Project management Company (PMC)	BECHTEL appointed by EMPLOYER
Supplier	An entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Sub- supplier	A sub entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Shall	Is to be understood as mandatory in relation to the requirements of this Document
Should	Is to be understood as a strong recommendation to comply with the requirements of this Document

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1. INTENT

This document is to Prepare for design of **Low Grade Thickening (3320)** Structure's Foundation. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2. SCOPE

The analysis has been carried out in Excel and Staad. For member strength calculation force values are directly taken from Load Data Provided & Output of Staad analysis.

3. DESIGN REQUIREMENTS

3.1 Design Load

All Design Loads are indicated in Foundation Load Plan (Doc. No. MD-216-3320-EG-ME-GAD- 1002) attached in Appendix A. Also indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

3.2 Load Combinations

Load combinations are based on SBC - 301.
Basic Load Cases are –

1. Tank Dead Load (D)
2. Max. Fluid Load (F1)
3. Normal Operational Fluid Load (F2)
4. Seismic Load (E)
5. Wind Load (W)

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Allowable Stress Design

Load Case	Load Combination
201	D + F1
202	D + F1 + W
203	D + F2+ 0.7 E
204	0.6 D + W
205	0.6 (D + F2) + 0.7 E

Ultimate Strength Design

Load Case	Load Combination
301	1.4 (D + F1)
302	1.2 (D + F1) + 1.6 W
303	1.2 (D + F2) + E
304	0.9 D + 1.6 W
305	0.9 (D+ F2) + E

3.3 Stability against Overturning & Sliding

Overturning

Structures and buildings foundations, shall have the following minimum stability ratios against overturning:

- 1.5 For load combinations with wind or seismic loads.
- 2.0 For all other load combinations.

Sliding

A minimum factor of safety against sliding shall be 1.5.

4. CONCRETE FOUNDATIONS

4.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304 and ACI 318M.

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4.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 MPa

Reinforcement Grade

- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

4.3 Foundations for Structures

Foundation shall be shallow isolated type. For design calculations as determined by the standard in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001).

4.4 Grouting

Grout shall be 30 mm thick cementitious under steel column bases of Strength 40 MPa.

4.5 Minimum Reinforcing Steel (Rebar) Size

The minimum rebar size for use as tension, compression reinforcement shall be 12 mm, except where welded wire fabric is used. For ties and spirals the minimum rebar size shall be 10 mm.

4.6 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Foundation and Column	50

4.7 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 450 kN/m^2 at 1.5m depth below EGL (ie. RL 910.50m) refer Table 5 - BH-26 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

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5. DESIGN CALCULATION FOR FOUNDATION

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DESIGN OF FOUNDATION MKD. F1

Size of Columns	Along-X	Along-Z	ex	ez
Outer Column A	0.65m	0.65m	0.0000m	-5.65m
2nd Middle Column B	0.60m	0.60m	0.0000m	-2.30m
1st Middle Column C	0.60m	0.60m	0.0000m	1.50m
Inner Column D	0.60m	0.60m	0.0000m	5.75m
Size of fdn. raft:	1.50m	13.00m		
Thickness of raft:	0.70 m			
Original Gr. Level:	0.00 m			
Finished Gr. Level:	0.00 m			
U/S of B/Pl Above GL:	2.10 m			
U/S level of fdn:	-1.50 m			
Unit weight of conc:	25 kN/m^3			
Unit weight of soil:	18 kN/m^3			

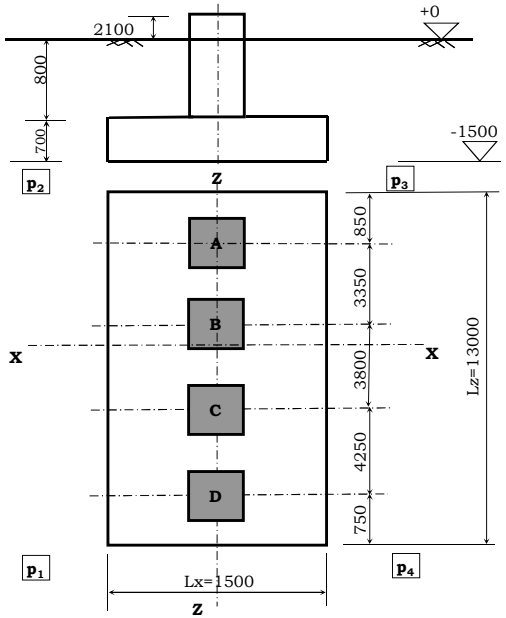
Calculation of Weight of Foundation :

Weight of raft = (1.5*13*0.7*25) = 341.25KN

Weight of soil = ((1.5*13)-2*(0.65*0.65))*0.8*18 = 268.63KN

Total = 609.88KN

Submerged wt. = 609.88-1.5*13*1.5*10 = 317.38KN



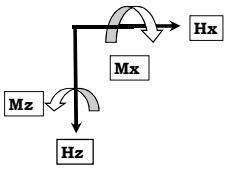
Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f$_y$=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	16mm
Dia. of bars provided along Z direction at top=	32mm
Mininum percentage of reinforcement at bottom=	0.18%
Mininum percentage of reinforcement at top=	0.09%

Effective depth of slab along X direction = 626mm = 0.626m

Effective depth of slab along Z direction = 634mm = 0.634m

Clear projection along X direction = 0.45m



H_X=Force along X-X direction

Hz=Force along Z-Z direction

M_X=Moment about X-X direction

Mz=Moment about Z-Z direction

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Basic Load Cases :

		Outer Column A		2nd Middle Column B		1st Middle Column C		Inner Column D		
Sl. No	Basic Load Case	V (kN)	Hx (kN)	V (kN)	Hx (kN)	V (kN)	Hx (kN)	V (kN)	Hx (kN)	Imcluding Bridge load 34+123=157kN
1	Tank Dead Load (D)	157.0	0.0	6.0	0.0	10.0	0.0	8.0	0.0	
2	Max. Fluid Load (F1)	680.0	112.0	822.0	0.0	784.0	0.0	582.0	0.0	
3	Normal Operational Fluid Load (F2)	560.0	93.0	677.0	0.0	646.0	0.0	479.0	0.0	
4	Wind Load (W)(+/-)	16.0	9.0	2.0	0.0	1.0	0.0	1.0	0.0	
5	Earthquake Load (E)(+/-)	44.0	26.0	3.0	0.0	0.0	0.0	0.0	0.0	

Combined Load Cases:		**Load at top of foundation**					**Load at bottom of foundation**				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
	Allowable Stress Design										
201	D + F1	3049.0	112.0	0.0	0.0	-2050.0	3658.88	112.00	403.20	0.00	-2049.95
202	D + F1 + W	3069.0	121.0	0.0	0.0	-2137.7	3678.88	121.00	435.60	0.00	-2137.70
203	D + F2+ 0.7 E	2575.9	111.2	0.0	0.0	-2016.6	3185.78	111.20	400.32	0.00	-2016.55
204	0.6 D + W	128.6	9.0	0.0	0.0	-591.7	738.48	9.00	32.40	0.00	-591.66
205	0.6(D+F2)+0.7 E	1558.7	74.0	0.0	0.0	-1281.5	2168.58	74.00	266.40	0.00	-1281.47
	Ultimate Conditions										
301	1.4 (D + F1)	4268.6	156.8	0.0	0.0	-2869.9	4878.48	156.80	564.48	0.00	-2869.93
302	1.2(D + F1) + 1.6 W	3690.8	148.8	0.0	0.0	-2547.7	4300.68	148.80	535.68	0.00	-2547.69
303	1.2(D + F2) + E	3098.6	137.6	0.0	0.0	-2460.7	3708.48	137.60	495.36	0.00	-2460.74
304	0.9 D + 1.6 W	194.9	14.4	0.0	0.0	-843.6	804.78	14.40	51.84	0.00	-843.62
305	0.9 (D+ F2) + E	2335.7	109.7	0.0	0.0	-1909.4	2945.58	109.70	394.92	0.00	-1909.43

Check for overturning and sliding:

Load Cases:		Load at b/o fdn					F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
		V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)				
201	D + F1	3658.88	112.00	403.20	0.00	2049.95	6.81	11.60	13.07	
202	D + F1 + W	3678.88	121.00	435.60	0.00	2137.70	6.33	11.19	12.16	
203	D + F2+ 0.7 E	3185.78	111.20	400.32	0.00	2016.55	5.97	10.27	11.46	
204	0.6 D + W	738.48	9.00	32.40	0.00	591.66	17.09	8.11	32.82	
205	0.6(D+F2)+0.7 E	2168.58	74.00	266.40	0.00	1281.47	6.11	11.00	11.72	

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DESIGN OF FOUNDATION MKD. F1

Load Combination & Calculation of Base pressure

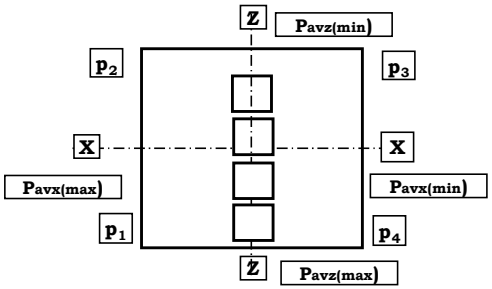
Max. Base pressure = p_{max} = (V/(Lx.Lz)) + (6.My/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)

Min. Base pressure = p_{min} = (V/(Lx.Lz)) - (6.My/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)

Surcharge load = 20 kN/m^2

Net safe bearing capacity of soil for DL, LL= 450 kN/m^2 For Wind load/Seis. load cas 562.5 kN/m^2

Gross safe bearing capacity of soil for DL, LL= 477 kN/m^2 For Wind load/Seis. load cas 589.5 kN/m^2



Load Case	Load Combination	Design Load/moments			Base Pressure (kN/m^2)												Remarks
		V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	
201	D + F1	3658.88	403.20	-2049.95	187.63	82.71	48.52	318.86	221.82	153.45	56.41	291.86	270.34	104.93	236.16	139.12	**O.K**
202	D + F1 + W	3678.88	435.60	-2137.70	188.66	89.35	50.60	328.61	227.42	149.90	48.71	301.61	278.02	99.31	239.26	138.07	**O.K**
203	D + F2+ 0.7 E	3185.78	400.32	-2016.55	163.37	82.12	47.73	293.22	197.76	128.99	33.53	266.22	245.49	81.26	211.11	115.65	**O.K**
204	0.6 D + W	738.48	32.40	-591.66	37.87	6.65	14.00	58.52	30.51	45.23	17.22	31.52	44.52	31.23	51.88	23.87	**O.K**
205	0.6(D+F2)+0.7 E	2168.58	266.40	-1281.47	111.21	54.65	30.33	196.19	135.52	86.89	26.23	169.19	165.86	56.56	141.54	80.88	**O.K**
301	1.4 (D + F1)	4878.48	564.48	-2869.93	250.18	115.79	67.93	433.90	298.04	202.32	66.46	406.90	365.97	134.39	318.11	182.25	
302	1.2(D + F1) + 1.6 W	4300.68	535.68	-2547.69	220.55	109.88	60.30	390.73	270.13	170.96	50.36	363.73	330.43	110.66	280.85	160.25	
303	1.2(D + F2) + E	3708.48	495.36	-2460.74	190.18	101.61	58.24	350.03	233.55	146.81	30.32	323.03	291.79	88.57	248.42	131.94	
304	0.9 D + 1.6 W	804.78	51.84	-843.62	41.27	10.63	19.97	71.87	31.94	50.60	10.67	44.87	51.91	30.64	61.24	21.31	
305	0.9 (D+ F2) + E	2945.58	394.92	-1909.43	151.06	81.01	45.19	277.26	186.87	115.24	24.85	250.26	232.07	70.05	196.25	105.86	

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DESIGN OF FOUNDATION MKD. F1

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.65+2*0.5*0.634)+(0.65+2*0.5*0.634))*2

= 5.136 m

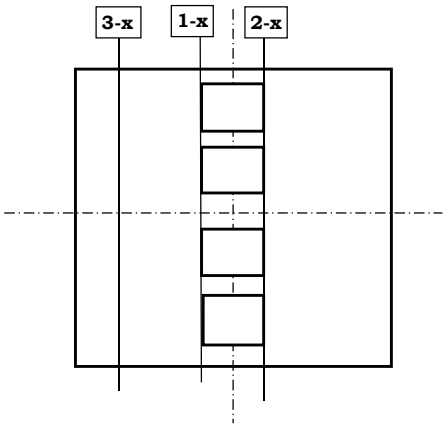
Punching Area= ((0.65+2*0.5*0.634)*(0.65+2*0.5*0.634))

= 1.65 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.7*25+0.8*0.7*18)

= 31.90 kN/m^2

Area of footing, A_f= 19.50 m^2



Load Case	Load Combination	p_{1x}	p_{2x}	p_{3x}	$B.M_{1x}$	$B.M_{2x}$	$S.F_{3x}$	$S.F_{(punching)}$
301	1.4 (D + F1)	296.50	203.86	393.14	31.48	12.72	-61.19	682.58
302	1.2(D + F1) + 1.6 W	264.50	176.59	356.22	28.00	10.20	-54.81	583.98
303	1.2(D + F2) + E	230.82	149.54	315.63	24.26	7.80	-47.84	523.54
304	0.9 D + 1.6 W	45.53	37.02	54.41	1.81	0.09	-3.74	82.18
305	0.9 (D+ F2) + E	183.46	118.66	251.08	18.63	5.50	-36.90	386.58

Overhang = 0.45 m

Effective depth = 0.626 m

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DESIGN OF FOUNDATION MKD. F1

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	31.48	0.08	0.11	0.000	159.93	1126.80	16ϕT	178.44	300	175	1148.93
At top	0.09	0.00	0.00	0.000	0.00	563.40	12ϕT	200.74	300	200	565.49

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= -3.74 kN/m

Shear stress , τ_v= (-3.74*1000)/(1000*626)

= -0.01 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 682.58 kN

Strength of Concrete in shear Vc1 = 9510.50 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 7034.12 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 6340.33 kN **Table 22.6.5.2_ (c)**

Design Vc= 6340.33 kN

Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ = (682.58*1000)/(5136*634)

= 0.21 N/mm^2 **No shear reinforcement reqd.**

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MD-216-3320-EG-CV-CAL-1210	Mansourah Massarah Gold Project	
900707_GAU01_ST02_CED01_ 33201_OU100665184_R3	DESIGN CALCULATION LOW GRADE THICKENING DESIGN OF FOUNDATION	Date: 15.11.2023
O19045_450_C_K_24088_001_R3	AS BUILT	Rev: D01

DESIGN OF FOUNDATION MKD. F1

Calculation of distribution reinf in Z direction :

Effective depth = 0.634 m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	118.94	0.30	0.39	0.001	599.58	1141.20	16ϕT	176.18	300	150	1340.41
At top	832.97	2.07	2.76	0.007	4385.00	570.60	32ϕT	183.41	300	175	4595.70

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 310.60 kN/m

Shear stress , τ_v= (310.60*1000)/(1000*634)

= 0.49 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

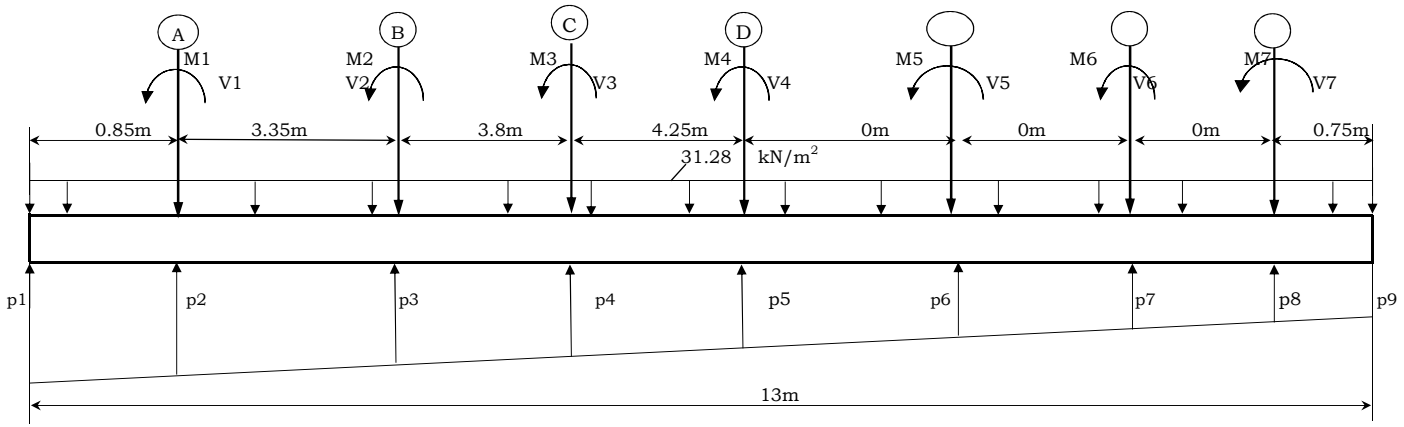
Along X Direction :			
At Bottom	**16ϕT**	**175**	c/c
At Top	**12ϕT**	**200**	c/c
Along Z Direction :			
At Bottom	**16ϕT**	**150**	c/c
At Top	**32ϕT**	**175**	c/c

MD-216-3320-EG-CV-CAL-1210	Mansourah Massarah Gold Project	
900707_GAU01_ST02_CED01_ 33201_OU100665184_R3	DESIGN CALCULATION LOW GRADE THICKENING DESIGN OF FOUNDATION	Date: 15.11.2023
O19045_450_C_K_24088_001_R3	AS BUILT	Rev: D01

DESIGN OF FOUNDATION MKD. F1

CALCULATION OF BM & SF :

(Along Z direction)



Load Case	Load Combination	Soil pressure in UDL (kN/m)								
		p1	p2	p3	p4	p5	p6	p7	p8	p9
301	1.4 (D + F1)	548.96	By STAAD							201.59
302	1.2(D + F1) + 1.6 W	495.65								165.99
303	1.2(D + F2) + E	437.69								132.85
304	0.9 D + 1.6 W	77.86								45.95
305	0.9 (D+ F2) + E	348.10								105.07

From STAAD Output	
BM at top	1249.46 (kNm/m)
BM at bot.	178.42 (kNm/m)
SF (T. at top)	423.33 (kN/m)
SF (T. at bottom)	465.90 (kN/m)

Load Case	Load Combination	Vertical Load (kN/m)							Moment (kNm/m)						
		V1	V2	V3	V4	V5	V6	V7	M1	M2	M3	M4	M5	M6	M7
301	1.4 (D + F1)	1171.80	1159.20	1111.60	826.00				564.48						
302	1.2(D + F1) + 1.6 W	1020.40	995.60	953.80	709.00				535.68						
303	1.2(D + F2) + E	904.40	822.60	787.20	584.40				495.36						
304	0.9 D + 1.6 W	157.30	7.40	10.00	8.20				51.84						
305	0.9 (D+ F2) + E	689.30	617.70	590.40	438.30				394.92						

Beam Force Detail

Sign convention as diagrams:- positive above line, negative below line except Fx where positive is compression. Distance d is given from beam end A.

			Axial	Shear		Torsion	Bending	
Beam	L/C	d (m)	Fx (kN)	Fy (kN)	Fz (kN)	Mx (kN-m)	My (kN-m)	Mz (kN-m)
1	301:1.4 (D + F	0.000	0.00	0.00	0.00	0.00	0.00	-0.00
		0.085	0.00	42.49	0.00	0.00	0.00	-2.01
		0.170	0.00	84.79	0.00	0.00	0.00	-7.52
		0.255	0.00	126.90	0.00	0.00	0.00	-16.52
		0.340	0.00	168.82	0.00	0.00	0.00	-28.99
		0.425	0.00	210.56	0.00	0.00	0.00	-44.91
		0.510	0.00	252.08	0.00	0.00	0.00	-64.77
		0.595	0.00	293.41	0.00	0.00	0.00	-88.06
		0.680	0.00	334.56	0.00	0.00	0.00	-114.75
		0.765	0.00	375.52	0.00	0.00	0.00	-144.83
		0.850	-0.00	416.29	-0.00	-0.00	-0.00	-178.29
	302:1.2(D + F1	0.000	0.00	0.00	0.00	0.00	0.00	-0.00
		0.085	0.00	-4.07	0.00	0.00	0.00	0.19
		0.170	0.00	-8.13	0.00	0.00	0.00	0.72
		0.255	0.00	-12.20	0.00	0.00	0.00	1.58
		0.340	0.00	-16.27	0.00	0.00	0.00	2.78
		0.425	0.00	-20.34	0.00	0.00	0.00	4.32
		0.510	0.00	-24.40	0.00	0.00	0.00	6.24
		0.595	0.00	-28.47	0.00	0.00	0.00	8.50
		0.680	0.00	-32.54	0.00	0.00	0.00	11.09
		0.765	0.00	-36.61	0.00	0.00	0.00	14.02
		0.850	-0.00	-40.67	-0.00	-0.00	-0.00	17.29
	303:1.2(D + F2	0.000	0.00	0.00	0.00	0.00	0.00	-0.00
		0.085	0.00	33.04	0.00	0.00	0.00	-1.56
		0.170	0.00	65.92	0.00	0.00	0.00	-5.85
		0.255	0.00	98.63	0.00	0.00	0.00	-12.84
		0.340	0.00	131.18	0.00	0.00	0.00	-22.53
		0.425	0.00	163.56	0.00	0.00	0.00	-34.91
		0.510	0.00	195.76	0.00	0.00	0.00	-50.33
		0.595	0.00	227.79	0.00	0.00	0.00	-68.41
		0.680	0.00	259.66	0.00	0.00	0.00	-89.13
		0.765	0.00	291.36	0.00	0.00	0.00	-112.47
		0.850	-0.00	322.89	-0.00	-0.00	-0.00	-138.43
	304:0.9 D + 1.6	0.000	0.00	0.00	0.00	0.00	0.00	-0.00
		0.085	0.00	-4.07	0.00	0.00	0.00	0.19
		0.170	0.00	-8.13	0.00	0.00	0.00	0.72
		0.255	0.00	-12.20	0.00	0.00	0.00	1.58
		0.340	0.00	-16.27	0.00	0.00	0.00	2.78
		0.425	0.00	-20.34	0.00	0.00	0.00	4.32
		0.510	0.00	-24.40	0.00	0.00	0.00	6.24
		0.595	0.00	-28.47	0.00	0.00	0.00	8.50
		0.680	0.00	-32.54	0.00	0.00	0.00	11.09
		0.765	0.00	-36.61	0.00	0.00	0.00	14.02
		0.850	-0.00	-40.67	-0.00	-0.00	-0.00	17.29
	305:0.9 (D+ F2	0.000	0.00	0.00	0.00	0.00	0.00	0.00
		0.085	0.00	25.45	0.00	0.00	0.00	-1.20
		0.170	0.00	50.76	0.00	0.00	0.00	-4.50

Beam Force Detail Cont...

			Axial	Shear		Torsion	Bending	
Beam	L/C	d	Fx	Fy	Fz	Mx	My	Mz
		(m)	(kN)	(kN)	(kN)	(kN-m)	(kN-m)	(kN-m)
		0.255	0.00	75.94	0.00	0.00	0.00	-9.89
		0.340	0.00	101.00	0.00	0.00	0.00	-17.35
		0.425	0.00	125.92	0.00	0.00	0.00	-26.88
		0.510	0.00	150.69	0.00	0.00	0.00	-38.75
		0.595	0.00	175.33	0.00	0.00	0.00	-52.67
		0.680	0.00	199.84	0.00	0.00	0.00	-68.61
		0.765	0.00	224.21	0.00	0.00	0.00	-86.58
		0.850	-0.00	248.46	-0.00	-0.00	-0.00	-106.55
2	301:1.4 (D + F	0.000	0.00	-862.88	0.00	0.00	0.00	386.19
		0.335	0.00	-704.28	0.00	0.00	0.00	645.66
		0.670	0.00	-548.60	0.00	0.00	0.00	853.98
		1.005	0.00	-395.84	0.00	0.00	0.00	1012.08
		1.340	0.00	-245.98	0.00	0.00	0.00	1120.89
		1.675	0.00	-99.05	0.00	0.00	0.00	1181.34
		2.010	0.00	44.55	0.00	0.00	0.00	1187.72
		2.345	0.00	185.24	0.00	0.00	0.00	1147.83
		2.680	0.00	323.02	0.00	0.00	0.00	1062.60
		3.015	0.00	457.87	0.00	0.00	0.00	932.97
		3.350	-0.00	589.81	-0.00	-0.00	-0.00	759.86
	302:1.2(D + F1	0.000	0.00	190.39	0.00	0.00	0.00	552.97
		0.335	0.00	174.36	0.00	0.00	0.00	492.17
		0.670	0.00	158.33	0.00	0.00	0.00	436.59
		1.005	0.00	142.30	0.00	0.00	0.00	386.24
		1.340	0.00	126.27	0.00	0.00	0.00	341.11
		1.675	0.00	110.24	0.00	0.00	0.00	301.19
		2.010	0.00	94.21	0.00	0.00	0.00	267.25
		2.345	0.00	78.18	0.00	0.00	0.00	238.52
		2.680	0.00	62.15	0.00	0.00	0.00	215.02
		3.015	0.00	46.12	0.00	0.00	0.00	196.73
		3.350	-0.00	30.09	-0.00	-0.00	-0.00	183.67
	303:1.2(D + F2	0.000	0.00	-679.74	0.00	0.00	0.00	356.93
		0.335	0.00	-557.28	0.00	0.00	0.00	561.78
		0.670	0.00	-437.38	0.00	0.00	0.00	727.20
		1.005	0.00	-320.04	0.00	0.00	0.00	853.98
		1.340	0.00	-205.26	0.00	0.00	0.00	942.96
		1.675	0.00	-93.04	0.00	0.00	0.00	994.95
		2.010	0.00	16.26	0.00	0.00	0.00	1005.70
		2.345	0.00	123.00	0.00	0.00	0.00	981.31
		2.680	0.00	227.19	0.00	0.00	0.00	922.57
		3.015	0.00	328.81	0.00	0.00	0.00	830.31
		3.350	-0.00	427.87	-0.00	-0.00	-0.00	705.34
	304:0.9 D + 1.6	0.000	0.00	60.33	0.00	0.00	0.00	69.13
		0.335	0.00	44.30	0.00	0.00	0.00	51.90
		0.670	0.00	28.27	0.00	0.00	0.00	39.89
		1.005	0.00	12.24	0.00	0.00	0.00	33.11
		1.340	0.00	-3.79	0.00	0.00	0.00	31.55
		1.675	0.00	-19.82	0.00	0.00	0.00	35.20
		2.010	0.00	-35.85	0.00	0.00	0.00	44.83
		2.345	0.00	-51.88	0.00	0.00	0.00	59.67

Beam Force Detail Cont...

			Axial	Shear		Torsion	Bending	
Beam	L/C	d (m)	Fx (kN)	Fy (kN)	Fz (kN)	Mx (kN-m)	My (kN-m)	Mz (kN-m)
		2.680	0.00	-67.91	0.00	0.00	0.00	79.74
		3.015	0.00	-83.94	0.00	0.00	0.00	105.02
		3.350	-0.00	-99.97	-0.00	-0.00	-0.00	135.53
	305:0.9 (D+ F	0.000	0.00	-522.53	0.00	0.00	0.00	288.37
		0.335	0.00	-428.44	0.00	0.00	0.00	445.85
		0.670	0.00	-336.38	0.00	0.00	0.00	573.04
		1.005	0.00	-246.37	0.00	0.00	0.00	670.58
		1.340	0.00	-158.39	0.00	0.00	0.00	739.14
		1.675	0.00	-72.45	0.00	0.00	0.00	779.35
		2.010	0.00	11.15	0.00	0.00	0.00	788.01
		2.345	0.00	92.72	0.00	0.00	0.00	769.79
		2.680	0.00	172.24	0.00	0.00	0.00	725.34
		3.015	0.00	249.73	0.00	0.00	0.00	655.32
		3.350	-0.00	325.17	-0.00	-0.00	-0.00	560.37
3	301:1.4 (D + F	0.000	0.00	-609.21	0.00	0.00	0.00	759.86
		0.380	0.00	-463.58	0.00	0.00	0.00	960.50
		0.760	0.00	-321.70	0.00	0.00	0.00	1108.08
		1.140	0.00	-183.57	0.00	0.00	0.00	1203.95
		1.520	0.00	-49.19	0.00	0.00	0.00	1249.46
		1.900	0.00	81.43	0.00	0.00	0.00	1245.98
		2.280	0.00	207.77	0.00	0.00	0.00	1188.25
		2.660	0.00	330.36	0.00	0.00	0.00	1084.58
		3.040	0.00	449.20	0.00	0.00	0.00	936.33
		3.420	0.00	564.28	0.00	0.00	0.00	744.85
		3.800	-0.00	675.62	-0.00	-0.00	-0.00	511.51
	302:1.2(D + F	0.000	0.00	178.11	0.00	0.00	0.00	183.67
		0.380	0.00	159.92	0.00	0.00	0.00	119.83
		0.760	0.00	141.74	0.00	0.00	0.00	62.70
		1.140	0.00	123.56	0.00	0.00	0.00	12.29
		1.520	0.00	105.38	0.00	0.00	0.00	-31.39
		1.900	0.00	87.19	0.00	0.00	0.00	-68.37
		2.280	0.00	69.01	0.00	0.00	0.00	-97.66
		2.660	0.00	50.83	0.00	0.00	0.00	-120.24
		3.040	0.00	32.64	0.00	0.00	0.00	-136.10
		3.420	0.00	14.46	0.00	0.00	0.00	-145.24
		3.800	-0.00	-3.72	-0.00	-0.00	-0.00	-147.66
	303:1.2(D + F	0.000	0.00	-431.00	0.00	0.00	0.00	705.34
		0.380	0.00	-322.16	0.00	0.00	0.00	846.04
		0.760	0.00	-216.62	0.00	0.00	0.00	947.18
		1.140	0.00	-114.37	0.00	0.00	0.00	1009.95
		1.520	0.00	-15.42	0.00	0.00	0.00	1035.54
		1.900	0.00	80.25	0.00	0.00	0.00	1025.15
		2.280	0.00	172.15	0.00	0.00	0.00	975.15
		2.660	0.00	260.76	0.00	0.00	0.00	891.85
		3.040	0.00	346.08	0.00	0.00	0.00	776.43
		3.420	0.00	428.11	0.00	0.00	0.00	630.08
		3.800	-0.00	506.84	-0.00	-0.00	-0.00	454.01
	304:0.9 D + 1.6	0.000	0.00	116.01	0.00	0.00	0.00	135.53
		0.380	0.00	97.83	0.00	0.00	0.00	95.28

Software licensed to LNT ECC DIV	Job No 3320	Sheet No 4	Rev
	Part		
Job Title Mansourah Massarah Gold Project	Ref		
	By AvG	Date 21-Jan-20	Chd SJM
Client Ma'aden	File F1.std	Date/Time 24-Mar-2020 15:32	

Beam Force Detail Cont...

			Axial	Shear		Torsion	Bending	
Beam	L/C	d (m)	Fx (kN)	Fy (kN)	Fz (kN)	Mx (kN-m)	My (kN-m)	Mz (kN-m)
		0.760	0.00	79.65	0.00	0.00	0.00	61.76
		1.140	0.00	61.46	0.00	0.00	0.00	34.94
		1.520	0.00	43.28	0.00	0.00	0.00	14.85
		1.900	0.00	25.10	0.00	0.00	0.00	1.48
		2.280	0.00	6.92	0.00	0.00	0.00	-4.22
		2.660	0.00	-11.27	0.00	0.00	0.00	-3.20
		3.040	0.00	-29.45	0.00	0.00	0.00	4.53
		3.420	0.00	-47.63	0.00	0.00	0.00	18.99
		3.800	-0.00	-65.82	-0.00	-0.00	-0.00	40.16
	305:0.9 (D+ F2	0.000	0.00	-322.38	0.00	0.00	0.00	560.37
		0.380	0.00	-239.62	0.00	0.00	0.00	665.33
		0.760	0.00	-159.48	0.00	0.00	0.00	740.22
		1.140	0.00	-81.97	0.00	0.00	0.00	786.00
		1.520	0.00	-7.09	0.00	0.00	0.00	803.62
		1.900	0.00	65.17	0.00	0.00	0.00	794.03
		2.280	0.00	134.43	0.00	0.00	0.00	754.56
		2.660	0.00	201.07	0.00	0.00	0.00	690.02
		3.040	0.00	265.08	0.00	0.00	0.00	601.36
		3.420	0.00	326.47	0.00	0.00	0.00	489.52
		3.800	-0.00	385.23	-0.00	-0.00	-0.00	355.46
4	301:1.4 (D + F	0.000	0.00	-399.19	0.00	0.00	0.00	511.51
		0.425	0.00	-279.75	0.00	0.00	0.00	652.81
		0.850	0.00	-165.01	0.00	0.00	0.00	745.78
		1.275	0.00	-54.95	0.00	0.00	0.00	792.33
		1.700	0.00	50.41	0.00	0.00	0.00	794.36
		2.125	0.00	151.08	0.00	0.00	0.00	753.76
		2.550	0.00	246.39	0.00	0.00	0.00	666.89
		2.975	0.00	337.00	0.00	0.00	0.00	541.67
		3.400	0.00	422.93	0.00	0.00	0.00	379.99
		3.825	0.00	504.16	0.00	0.00	0.00	183.76
		4.250	-0.00	580.70	-0.00	-0.00	-0.00	-45.12
	302:1.2(D + F1	0.000	0.00	63.77	0.00	0.00	0.00	-147.66
		0.425	0.00	43.43	0.00	0.00	0.00	-169.96
		0.850	0.00	23.10	0.00	0.00	0.00	-183.86
		1.275	0.00	2.76	0.00	0.00	0.00	-189.36
		1.700	0.00	-17.57	0.00	0.00	0.00	-186.45
		2.125	0.00	-37.91	0.00	0.00	0.00	-175.14
		2.550	0.00	-58.25	0.00	0.00	0.00	-154.23
		2.975	0.00	-78.58	0.00	0.00	0.00	-124.91
		3.400	0.00	-98.92	0.00	0.00	0.00	-87.19
		3.825	0.00	-119.26	0.00	0.00	0.00	-41.07
		4.250	-0.00	-139.59	-0.00	-0.00	-0.00	13.46
	303:1.2(D + F2	0.000	0.00	-246.34	0.00	0.00	0.00	454.01
		0.425	0.00	-162.74	0.00	0.00	0.00	538.82
		0.850	0.00	-83.26	0.00	0.00	0.00	590.00
		1.275	0.00	-7.89	0.00	0.00	0.00	609.20
		1.700	0.00	63.36	0.00	0.00	0.00	598.10
		2.125	0.00	130.49	0.00	0.00	0.00	558.35
		2.550	0.00	192.91	0.00	0.00	0.00	488.02

Beam Force Detail Cont...

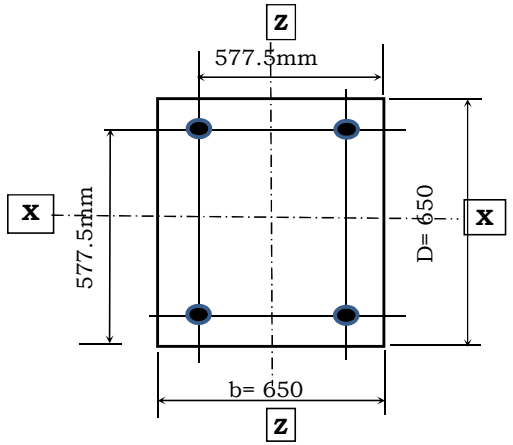
			Axial	Shear		Torsion	Bending	
Beam	L/C	d (m)	Fx (kN)	Fy (kN)	Fz (kN)	Mx (kN-m)	My (kN-m)	Mz (kN-m)
		2.975	0.00	251.22	0.00	0.00	0.00	392.79
		3.400	0.00	305.40	0.00	0.00	0.00	274.35
		3.825	0.00	355.47	0.00	0.00	0.00	134.34
		4.250	-0.00	401.43	-0.00	-0.00	-0.00	-25.56
	304:0.9 D + 1.6	0.000	0.00	107.96	0.00	0.00	0.00	40.16
		0.425	0.00	87.63	0.00	0.00	0.00	-0.92
		0.850	0.00	67.29	0.00	0.00	0.00	-33.61
		1.275	0.00	46.95	0.00	0.00	0.00	-57.88
		1.700	0.00	26.62	0.00	0.00	0.00	-73.76
		2.125	0.00	6.28	0.00	0.00	0.00	-81.23
		2.550	0.00	-14.05	0.00	0.00	0.00	-79.10
		2.975	0.00	-34.39	0.00	0.00	0.00	-68.56
		3.400	0.00	-54.73	0.00	0.00	0.00	-49.63
		3.825	0.00	-75.06	0.00	0.00	0.00	-22.29
		4.250	-0.00	-95.40	-0.00	-0.00	-0.00	13.46
	305:0.9 (D+ F2	0.000	0.00	-176.21	0.00	0.00	0.00	355.46
		0.425	0.00	-114.04	0.00	0.00	0.00	415.56
		0.850	0.00	-55.16	0.00	0.00	0.00	450.69
		1.275	0.00	0.45	0.00	0.00	0.00	462.18
		1.700	0.00	52.77	0.00	0.00	0.00	451.37
		2.125	0.00	101.80	0.00	0.00	0.00	419.57
		2.550	0.00	147.09	0.00	0.00	0.00	365.50
		2.975	0.00	189.09	0.00	0.00	0.00	293.44
		3.400	0.00	227.81	0.00	0.00	0.00	204.71
		3.825	0.00	263.25	0.00	0.00	0.00	100.66
		4.250	-0.00	295.40	-0.00	-0.00	-0.00	-17.41
5	301:1.4 (D + F	0.000	0.00	-122.82	0.00	0.00	0.00	-45.12
		0.075	0.00	-109.87	0.00	0.00	0.00	-36.45
		0.150	0.00	-97.07	0.00	0.00	0.00	-28.71
		0.225	0.00	-84.41	0.00	0.00	0.00	-21.91
		0.300	0.00	-71.90	0.00	0.00	0.00	-16.02
		0.375	0.00	-59.53	0.00	0.00	0.00	-11.04
		0.450	0.00	-47.33	0.00	0.00	0.00	-7.09
		0.525	0.00	-35.28	0.00	0.00	0.00	-4.02
		0.600	0.00	-23.37	0.00	0.00	0.00	-1.82
		0.675	0.00	-11.61	0.00	0.00	0.00	-0.48
		0.750	-0.00	0.00	-0.00	-0.00	-0.00	-0.00
	302:1.2(D + F1	0.000	0.00	35.89	0.00	0.00	0.00	13.46
		0.075	0.00	32.30	0.00	0.00	0.00	10.92
		0.150	0.00	28.71	0.00	0.00	0.00	8.64
		0.225	0.00	25.12	0.00	0.00	0.00	6.62
		0.300	0.00	21.53	0.00	0.00	0.00	4.86
		0.375	0.00	17.94	0.00	0.00	0.00	3.36
		0.450	0.00	14.35	0.00	0.00	0.00	2.17
		0.525	0.00	10.77	0.00	0.00	0.00	1.23
		0.600	0.00	7.18	0.00	0.00	0.00	0.56
		0.675	0.00	3.59	0.00	0.00	0.00	0.15
		0.750	-0.00	-0.00	-0.00	-0.00	-0.00	0.00
	303:1.2(D + F2	0.000	0.00	-70.35	0.00	0.00	0.00	-25.56

Beam Force Detail Cont...

			Axial	Shear		Torsion	Bending	
Beam	L/C	d	Fx	Fy	Fz	Mx	My	Mz
		(m)	(kN)	(kN)	(kN)	(kN-m)	(kN-m)	(kN-m)
		0.075	0.00	-62.72	0.00	0.00	0.00	-20.60
		0.150	0.00	-55.23	0.00	0.00	0.00	-16.19
		0.225	0.00	-47.87	0.00	0.00	0.00	-12.33
		0.300	0.00	-40.63	0.00	0.00	0.00	-8.99
		0.375	0.00	-33.52	0.00	0.00	0.00	-6.18
		0.450	0.00	-26.56	0.00	0.00	0.00	-3.96
		0.525	0.00	-19.73	0.00	0.00	0.00	-2.24
		0.600	0.00	-13.02	0.00	0.00	0.00	-1.01
		0.675	0.00	-6.45	0.00	0.00	0.00	-0.27
		0.750	-0.00	-0.00	-0.00	-0.00	-0.00	0.00
	304:0.9 D + 1.6	0.000	0.00	35.89	0.00	0.00	0.00	13.46
		0.075	0.00	32.30	0.00	0.00	0.00	10.92
		0.150	0.00	28.71	0.00	0.00	0.00	8.64
		0.225	0.00	25.12	0.00	0.00	0.00	6.62
		0.300	0.00	21.53	0.00	0.00	0.00	4.86
		0.375	0.00	17.94	0.00	0.00	0.00	3.36
		0.450	0.00	14.36	0.00	0.00	0.00	2.17
		0.525	0.00	10.77	0.00	0.00	0.00	1.23
		0.600	0.00	7.18	0.00	0.00	0.00	0.56
		0.675	0.00	3.59	0.00	0.00	0.00	0.15
		0.750	-0.00	-0.00	-0.00	-0.00	-0.00	0.00
	305:0.9 (D+ F4	0.000	0.00	-48.17	0.00	0.00	0.00	-17.41
		0.075	0.00	-42.89	0.00	0.00	0.00	-14.02
		0.150	0.00	-37.71	0.00	0.00	0.00	-11.00
		0.225	0.00	-32.63	0.00	0.00	0.00	-8.37
		0.300	0.00	-27.65	0.00	0.00	0.00	-6.10
		0.375	0.00	-22.77	0.00	0.00	0.00	-4.19
		0.450	0.00	-18.01	0.00	0.00	0.00	-2.68
		0.525	0.00	-13.36	0.00	0.00	0.00	-1.51
		0.600	0.00	-8.80	0.00	0.00	0.00	-0.68
		0.675	0.00	-4.35	0.00	0.00	0.00	-0.18
		0.750	-0.00	-0.00	-0.00	-0.00	-0.00	0.00

MD-216-3320-EG-CV-CAL-1210	Mansourah Massarah Gold Project	
900707_GAU01_ST02_CED01_ 33201_OU100665184_R3	DESIGN CALCULATION LOW GRADE THICKENING DESIGN OF FOUNDATION	Date: 15.11.2023
O19045_450_C_K_24088_001_R3	AS BUILT	Rev: D01

DESIGN OF COLUMN MKD C1 **(Bridge/Drive in Outer Column)**



Input:

Clear cover to reinf.=	50mm	Table 20.6.1.3.1
Dia. of main reinf.=	25mm	
Dia. of link=	10mm	
Effective cover =	72.5mm	
Grade of Concrete (f_c') =	M35	
Grade of Steel (f_y) =	Fy420	
Grade of Reinf(links).=	Fy420	
Min. Steel Ratio	0.50%	
Max. Steel Ratio	4.00%	

For Compression-controlled sections

Strength reduction factor ϕ = 0.65 Table 21.2.2

Unsupproted length of column about X (l_{ux}) =	3050mm
Unsupproted length of column about Z (l_{uz}) =	3050mm

For Columns not braced against sidesway,

Where, Effective length factor K= 1.2

Slenderness ratio,	$K\,l_{ux} / r_x$ =	19.51	< 22 SECTION 6.2.5a	Radius of gyration r_x=	188	mm
	$K\,l_{uz} / r_z$ =	19.51	< 22 SECTION 6.2.5a	Radius of gyration r_z=	188	mm

MD-216-3320-EG-CV-CAL-1210	Mansourah Massarah Gold Project	
900707_GAU01_ST02_CED01_ 33201_OU100665184_R3	DESIGN CALCULATION LOW GRADE THICKENING DESIGN OF FOUNDATION	Date: 15.11.2023
O19045_450_C_K_24088_001_R3	AS BUILT	Rev: D01

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)	
1	Tank Dead Load (D)	324.0	0.0	0.0	0.0	0.0	Including Bridge DL+LL, V=(34+123+167)=324kN
2	Max. Fluid Load (F1)	680.0	112.0	0.0	0.0	0.0	
3	Normal Opl Fluid Load (F2)	560.0	93.0	0.0	0.0	0.0	
4	Wind Load (W)(+/-)	16.0	9.0	0.0	0.0	0.0	
5	Earthquake Load (E)(+/-)	44.0	26.0	0.0	0.0	0.0	
			γ_y =	0.81			
			γ_z =	0.81			

Load Case	Load Combination	Factored Load/Moment			Nominal Load/Moment			Kn = Pn/f'_cbD (N/mm^2)
		Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
301	1.4 (D + F1)	140.56	47.8	0.0	216.2	73.6	0.0	0.15
302	1.2(D + F1) + 1.6 W	123.0	45.4	0.0	189.3	69.8	0.0	0.13
303	1.2(D + F2) + E	124.9	42.0	0.0	192.1	64.6	0.0	0.13
304	0.9 D + 1.6 W	31.7	4.4	0.0	48.8	6.8	0.0	0.03
305	0.9 (D+ F2) + E	84.0	33.5	0.0	129.2	51.5	0.0	0.09

Reinforcement: Provide,

4-Nos	25 mm Dia
4-Nos	25 mm Dia
4-Nos	25 mm Dia

1964.29 Sq mm (At Corner)
1964.29 Sq mm (Along Z axis)
1964.29 Sq mm (Along Y axis)

A_{sc} pvd. = 5892.86 Sq mm

Total A_{sc} = 1.4 %

β = P_n / P_0 (range is 0.55 to 0.70)

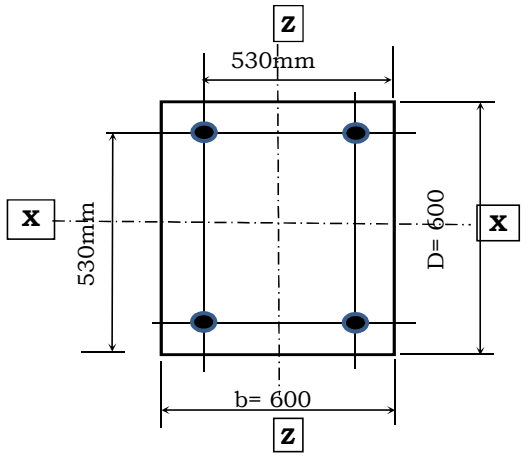
Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 1486.906 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.8		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f'_cb^2D$	$R_{nx}=M_{nox}/f'_cb^2D$						
301	0.10	0.10	96.1	96.1	0.55	1.16	0.73	O.K
302	0.09	0.09	86.5	86.5	0.55	1.16	0.78	O.K
303	0.09	0.09	86.5	86.5	0.55	1.16	0.71	O.K
304	0.05	0.05	48.1	48.1	0.55	1.16	0.10	O.K
305	0.07	0.07	67.3	67.3	0.55	1.16	0.73	O.K

Links, 10mm dia @ 300 C/C

MD-216-3320-EG-CV-CAL-1210	Mansourah Massarah Gold Project	
900707_GAU01_ST02_CED01_ 33201_OU100665184_R3	DESIGN CALCULATION LOW GRADE THICKENING DESIGN OF FOUNDATION	Date: 15.11.2023
O19045_450_C_K_24088_001_R3	AS BUILT	Rev: D01

DESIGN OF COLUMN MKD C2 **(2nd Middle Columns)**



Input:

Clear cover to reinf.=	50mm	Table 20.6.1.3.1
Dia. of main reinf.=	20mm	
Dia. of link=	10mm	
Effective cover =	70mm	
Grade of Concrete (f_c') =	M35	
Grade of Steel (f_y) =	Fy420	
Grade of Reinf(links).=	Fy420	
Min. Steel Ratio	0.50%	
Max. Steel Ratio	4.00%	

For Compression-controlled sections

Strength reduction factor ϕ= 0.65 Table 21.2.2

Unsupproted length of column about X (l_{ux}) =	3050mm
Unsupproted length of column about Z (l_{uz}) =	3050mm

For Columns not braced against sidesway, Where, Effective length factor K= 1.2

Slenderness ratio,	$K\,l_{ux} / r_x$ =	21.14	< 22 SECTION 6.2.5a	Radius of gyration r_x=	173	mm
	$K\,l_{uz} / r_z$ =	21.14	< 22 SECTION 6.2.5a	Radius of gyration r_z=	173	mm

MD-216-3320-EG-CV-CAL-1210	Mansourah Massarah Gold Project	
900707_GAU01_ST02_CED01_ 33201_OU100665184_R3	DESIGN CALCULATION LOW GRADE THICKENING DESIGN OF FOUNDATION	Date: 15.11.2023
O19045_450_C_K_24088_001_R3	AS BUILT	Rev: D01

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	Tank Dead Load (D)	6.0	0.0	0.0	0.0	0.0
2	Max. Fluid Load (F1)	822.0	0.0	0.0	0.0	0.0
3	Normal Opl Fluid Load (F2)	677.0	0.0	0.0	0.0	0.0
4	Wind Load (W)(+/-)	2.0	0.0	0.0	0.0	0.0
5	Earthquake Load (E)(+/-)	3.0	0.0	0.0	0.0	0.0
			γ_y =	0.80		
			γ_z =	0.80		

Load Case	Load Combination	Factored Load/Moment			Nominal Load/Moment			Kn = Pn/f'_cbD (N/mm^2)
		Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
301	1.4 (D + F1)	115.92	0.0	0.0	178.3	0.0	0.0	0.14
302	1.2(D + F1) + 1.6 W	99.7	0.0	0.0	153.4	0.0	0.0	0.12
303	1.2(D + F2) + E	99.7	0.0	0.0	153.3	0.0	0.0	0.12
304	0.9 D + 1.6 W	0.9	0.0	0.0	1.3	0.0	0.0	0.00
305	0.9 (D+ F2) + E	61.8	0.0	0.0	95.0	0.0	0.0	0.08

Reinforcement: Provide,

4-Nos	20 mm Dia
4-Nos	20 mm Dia
4-Nos	20 mm Dia

1257.14 Sq mm (At Corner)
1257.14 Sq mm (Along Z axis)
1257.14 Sq mm (Along Y axis)

A_{sc} pvd. = 3771.43 Sq mm

Total A_{sc} = 1.0 %

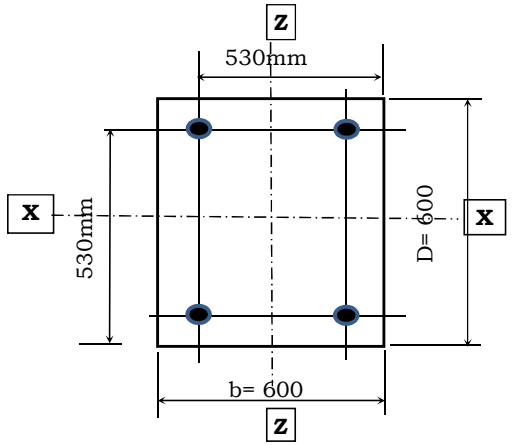
β = P_n / P_0 (range is 0.55 to 0.70) Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 1218.18 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.8		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f_cb^2D$	$R_{nx}=M_{nox}/f_cb^2D$						
301	0.06	0.06	45.4	45.4	0.55	1.16	0.00	O.K
302	0.06	0.06	45.4	45.4	0.55	1.16	0.00	O.K
303	0.06	0.06	45.4	45.4	0.55	1.16	0.00	O.K
304	0.03	0.03	22.7	22.7	0.55	1.16	0.00	O.K
305	0.04	0.04	30.2	30.2	0.55	1.16	0.00	O.K

Links, 10mm dia @ 300 C/C

MA'ADEN معادن		Outotec / LARSEN & TOUBRO
MD-216-3320-EG-CV-CAL-1210	Mansourah Massarah Gold Project	
900707_GAU01_ST02_CED01_ 33201_OU100665184_R3	DESIGN CALCULATION LOW GRADE THICKENING DESIGN OF FOUNDATION	Date: 15.11.2023
O19045_450_C_K_24088_001_R3	AS BUILT	Rev: D01

DESIGN OF COLUMN MKD C3 (1tst Middle Columns)



Input:

Clear cover to reinf.=	50mm	Table 20.6.1.3.1
Dia. of main reinf.=	20mm	
Dia. of link=	10mm	
Effective cover =	70mm	
Grade of Concrete (f_c') =	M35	
Grade of Steel (f_y) =	Fy420	
Grade of Reinf(links).=	Fy420	
Min. Steel Ratio	0.50%	
Max. Steel Ratio	4.00%	

For Compression-controlled sections

Strength reduction factor ϕ = 0.65 Table 21.2.2

Unsupproted length of column about X (l_{ux}) =	3050mm
Unsupproted length of column about Z (l_{uz}) =	3050mm

For Columns not braced against sideway, Where, Effective length factor K= 1.2

Slenderness ratio,	K l_{ux} / r_x =	21.14	< 22 SECTION 6.2.5a	Radius of gyration r_x=	173	mm
	K l_{uz} / r_z =	21.14	< 22 SECTION 6.2.5a	Radius of gyration r_z=	173	mm

MD-216-3320-EG-CV-CAL-1210	Mansourah Massarah Gold Project	
900707_GAU01_ST02_CED01_ 33201_OU100665184_R3	DESIGN CALCULATION LOW GRADE THICKENING DESIGN OF FOUNDATION	Date: 15.11.2023
O19045_450_C_K_24088_001_R3	AS BUILT	Rev: D01

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	Tank Dead Load (D)	10.0	0.0	0.0	0.0	0.0
2	Max. Fluid Load (F1)	784.0	0.0	0.0	0.0	0.0
3	Normal Opl Fluid Load (F2)	646.0	0.0	0.0	0.0	0.0
4	Wind Load (W)(+/-)	1.0	0.0	0.0	0.0	0.0
5	Earthquake Load (E)(+/-)	0.0	0.0	0.0	0.0	0.0
			γ_y =	0.80		
			γ_z =	0.80		

Load Case	Load Combination	Factored Load/Moment			Nominal Load/Moment			Kn = $P_n/f'_c bD$ (N/mm^2)
		Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
301	1.4 (D + F1)	111.16	0.0	0.0	171.0	0.0	0.0	0.14
302	1.2(D + F1) + 1.6 W	95.4	0.0	0.0	146.8	0.0	0.0	0.12
303	1.2(D + F2) + E	95.3	0.0	0.0	146.6	0.0	0.0	0.12
304	0.9 D + 1.6 W	1.1	0.0	0.0	1.6	0.0	0.0	0.00
305	0.9 (D+ F2) + E	59.0	0.0	0.0	90.8	0.0	0.0	0.07

Reinforcement: Provide,

4-Nos	20 mm Dia	1257.14 Sq mm (At Corner)	A_{sc} pvd. = 3771.43 Sq mm
4-Nos	20 mm Dia	1257.14 Sq mm (Along Z axis)	
4-Nos	20 mm Dia	1257.14 Sq mm (Along Y axis)	Total A_{sc} = 1.0 %

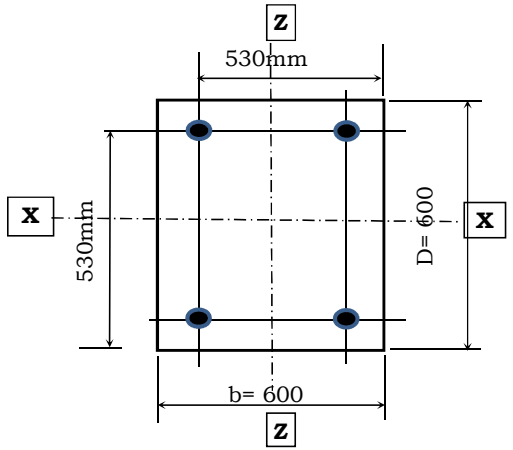
β = P_n / P_0 (range is 0.55 to 0.70) Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 1218.18 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.8		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f'_c b^2 D$	$R_{nx}=M_{nox}/f'_c b^2 D$						
301	0.06	0.06	45.4	45.4	0.55	1.16	0.00	O.K
302	0.06	0.06	45.4	45.4	0.55	1.16	0.00	O.K
303	0.06	0.06	45.4	45.4	0.55	1.16	0.00	O.K
304	0.03	0.03	22.7	22.7	0.55	1.16	0.00	O.K
305	0.05	0.05	37.8	37.8	0.55	1.16	0.00	O.K

Links, 10mm dia @ 300 C/C

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DESIGN OF COLUMN MKD C4 **(Inner Columns)**



Input:

Clear cover to reinf.=	50mm	Table 20.6.1.3.1
Dia. of main reinf.=	20mm	
Dia. of link=	10mm	
Effective cover =	70mm	
Grade of Concrete (f_c') =	M35	
Grade of Steel (f_y) =	Fy420	
Grade of Reinf(links).=	Fy420	
Min. Steel Ratio	0.50%	
Max. Steel Ratio	4.00%	

For Compression-controlled sections
Strength reduction factor ϕ= 0.65 Table 21.2.2

Unsupproted length of column about X (l_{ux}) =	3050mm
Unsupproted length of column about Z (l_{uz}) =	3050mm

For Columns not braced against sidesway, Where, Effective length factor K= 1.2

Slenderness ratio,	$K\,l_{ux} / r_x$ =	21.14	< 22 SECTION 6.2.5a	Radius of gyration r_x=	173	mm
	$K\,l_{uz} / r_z$ =	21.14	< 22 SECTION 6.2.5a	Radius of gyration r_z=	173	mm

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Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	Tank Dead Load (D)	8.0	0.0	0.0	0.0	0.0
2	Max. Fluid Load (F1)	582.0	0.0	0.0	0.0	0.0
3	Normal Opl Fluid Load (F2)	479.0	0.0	0.0	0.0	0.0
4	Wind Load (W)(+/-)	1.0	0.0	0.0	0.0	0.0
5	Earthquake Load (E)(+/-)	0.0	0.0	0.0	0.0	0.0

γ_y = 0.80

γ_z = 0.80

Load Case	Load Combination	Factored Load/Moment			Nominal Load/Moment			Kn = Pn/f'_cbD (N/mm^2)
		Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
301	1.4 (D + F1)	82.60	0.0	0.0	127.1	0.0	0.0	0.10
302	1.2(D + F1) + 1.6 W	71.0	0.0	0.0	109.2	0.0	0.0	0.09
303	1.2(D + F2) + E	70.8	0.0	0.0	108.9	0.0	0.0	0.09
304	0.9 D + 1.6 W	0.9	0.0	0.0	1.4	0.0	0.0	0.00
305	0.9 (D+ F2) + E	43.8	0.0	0.0	67.4	0.0	0.0	0.05

Reinforcement: Provide,

4-Nos	20 mm Dia
4-Nos	20 mm Dia
4-Nos	20 mm Dia

1257.14 Sq mm (At Corner)
1257.14 Sq mm (Along Z axis)
1257.14 Sq mm (Along Y axis)

A_{sc} pvd. = 3771.43 Sq mm

Total A_{sc} = 1.0 %

β = P_n / P_0 (range is 0.55 to 0.70) Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 1218.18 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.8		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f'_cb^2D$	$R_{nx}=M_{nox}/f'_cb^2D$						
301	0.06	0.06	45.4	45.4	0.55	1.16	0.00	O.K
302	0.06	0.06	45.4	45.4	0.55	1.16	0.00	O.K
303	0.06	0.06	45.4	45.4	0.55	1.16	0.00	O.K
304	0.03	0.03	22.7	22.7	0.55	1.16	0.00	O.K
305	0.04	0.04	30.2	30.2	0.55	1.16	0.00	O.K

Links, 10mm dia @ 300 C/C

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DESIGN OF FOUNDATION MKD. F2 (Central Column)

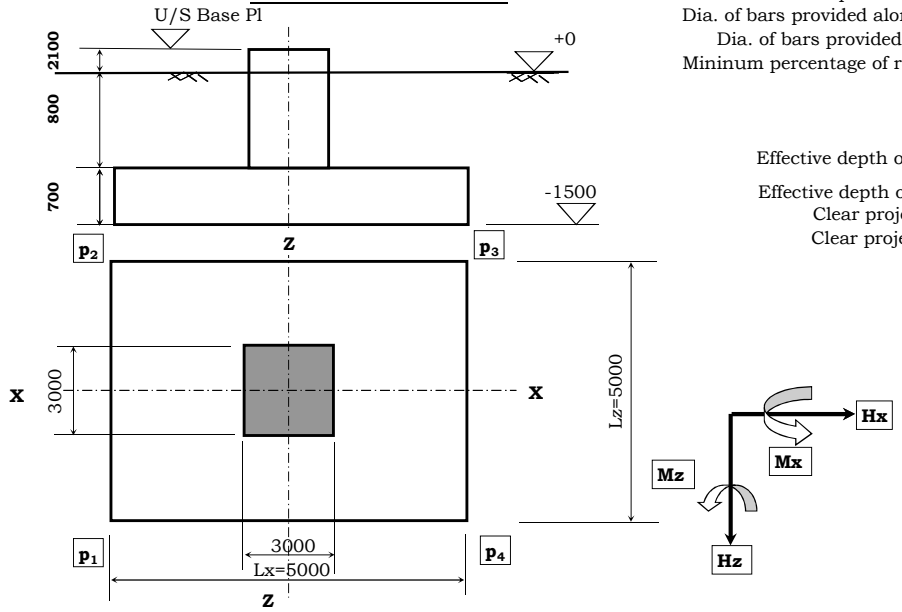
	Along-X	Along-Z
Size of column :	3.00m	3.00m
Size of fdn. raft :	5.00m	5.00m
Thickness of raft :	0.70 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation	-1.50 m	
U/S level Base Plate :	2.10 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

SYMBOLS AND NOTATIONS

Tank Dead Load	D
Max. Fluid Load	F1
Normal Operational Fluid Load	F2
Wind Load	W
Earthquake Load	E

Calculation of Weight of Foundation :

Weight of raft & Col.= (5*5*0.7*24)+(3*3*2.9*24) = 1046.40KN
Weight of soil = ((5*5)-(3*3))*0.8*18 = 230.40KN
Total = 1276.80KN



Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	20mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	20mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.18%

Effective depth of slab along X direction = 640mm = 0.64m
Effective depth of slab along Z direction = 620mm = 0.62m
Clear projection along Z direction = 1m
Clear projection along X direction = 1m

H_x=Force along X-X direction
Hz=Force along Z-Z direction
M_x=Moment about X-X direction
Mz=Moment about Z-Z direction

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DESIGN OF FOUNDATION MKD. F2

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mz (kNm)
1	Tank Dead Load (D)	152.0	0.0	0.0	0.0	0.0
2	Max. Fluid Load (F1)	8715.0	0.0	0.0	0.0	0.0
3	Normal Operational Fluid Load (F2)	7176.0	0.0	0.0	0.0	0.0
4	Wind Load (W)(+/-)	0.0	54.0	0.0	110.0	0.0
5	Earthquake Load (E)(+/-)	0.0	270.0	0.0	546.0	0.0

Combined Load Cases:		**Load at top of foundation**					**Load at bottom of foundation**				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
201	D + F1	8867.0	0.0	0.0	0.0	0.0	10143.80	0.00	0.00	0.00	0.00
202	D + F1 + W	8867.0	54.0	110.0	0.0	0.0	10143.80	54.00	304.40	0.00	0.00
203	D + F2+ 0.7 E	7328.0	189.0	382.2	0.0	0.0	8604.80	189.00	1062.60	0.00	0.00
204	0.6 D + W	91.2	54.0	110.0	0.0	0.0	1368.00	54.00	304.40	0.00	0.00
205	0.6(D+F2)+0.7 E	4396.8	189.0	382.2	0.0	0.0	5673.60	189.00	1062.60	0.00	0.00
Ultimate Conditions											
301	1.4 (D + F1)	12413.8	0.0	0.0	0.0	0.0	13690.60	0.00	0.00	0.00	0.00
302	1.2(D + F1) + 1.6 W	10640.4	86.4	176.0	0.0	0.0	11917.20	86.40	487.04	0.00	0.00
303	1.2(D + F2) + E	10640.4	270.0	546.0	0.0	0.0	11917.20	270.00	1518.00	0.00	0.00
304	0.9 D + 1.6 W	136.8	86.4	176.0	0.0	0.0	1413.60	86.40	487.04	0.00	0.00
305	0.9 (D+ F2) + E	6595.2	270.0	546.0	0.0	0.0	7872.00	270.00	1518.00	0.00	0.00

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Check for overturning and sliding:

Load Cases:		Load at b/o fdn					F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
		V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)				
201	D + F1	10143.80	0.00	0.00	0.00	0.00				
202	D + F1 + W	10143.80	54.00	304.40	0.00	0.00	83.31		67.63	
203	D + F2+ 0.7 E	8604.80	189.00	1062.60	0.00	0.00	20.24		16.39	
204	0.6 D + W	1368.00	54.00	304.40	0.00	0.00	11.24		9.12	
205	0.6(D+F2)+0.7 E	5673.60	189.00	1062.60	0.00	0.00	13.35		10.81	

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DESIGN OF FOUNDATION MKD. F2

Load Combination & Calculation of Base pressure

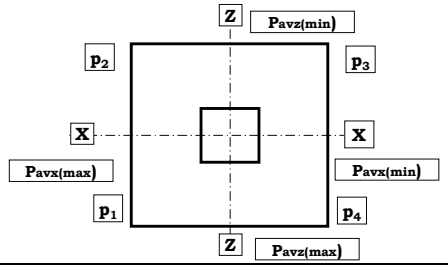
Max. Base pressure = p_{max} = (V/(Lx.Lz)) + (6.My/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)

Min. Base pressure = p_{min} = (V/(Lx.Lz)) - (6.My/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)

Surcharge load = 20 kN/m^2

Net SBC of soil for DL, LL= 450 kN/m^2 For WL/SL case = 562.5 kN/m^2

Gross SBC of soil for DL, LL= 477 kN/m^2 For WL/SL case = 589.5 kN/m^2



Load Case	Load Combination	Design Load/moments			Base Pressure (kN/m^2)												Remarks
		V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	
201	D + F1	10143.80	0.00	0.00	405.75	0.00	0.00	405.75	405.75	405.75	405.75	378.75	405.75	405.75	405.75	405.75	**O.K**
202	D + F1 + W	10143.80	304.40	0.00	405.75	14.61	0.00	420.36	420.36	391.14	391.14	393.36	420.36	391.14	405.75	405.75	**O.K**
203	D + F2+ 0.7 E	8604.80	1062.60	0.00	344.19	51.00	0.00	395.20	395.20	293.19	293.19	368.20	395.20	293.19	344.20	344.20	**O.K**
204	0.6 D + W	1368.00	304.40	0.00	54.72	14.61	0.00	69.33	69.33	40.11	40.11	42.33	69.33	40.11	54.72	54.72	**O.K**
205	0.6(D+F2)+0.7 E	5673.60	1062.60	0.00	226.94	51.00	0.00	277.95	277.95	175.94	175.94	250.95	277.95	175.94	226.95	226.95	**O.K**
301	1.4 (D + F1)	13690.60	0.00	0.00	547.62	0.00	0.00	547.62	547.62	547.62	547.62	520.62	547.62	547.62	547.62	547.62	
302	1.2(D + F1) + 1.6 W	11917.20	487.04	0.00	476.69	23.38	0.00	500.07	500.07	453.31	453.31	473.07	500.07	453.31	476.69	476.69	
303	1.2(D + F2) + E	11917.20	1518.00	0.00	476.69	72.86	0.00	549.55	549.55	403.82	403.82	522.55	549.55	403.82	476.69	476.69	A02
304	0.9 D + 1.6 W	1413.60	487.04	0.00	56.54	23.38	0.00	79.92	79.92	33.17	33.17	52.92	79.92	33.17	56.55	56.55	
305	0.9 (D+ F2) + E	7872.00	1518.00	0.00	314.88	72.86	0.00	387.74	387.74	242.02	242.02	360.74	387.74	242.02	314.88	314.88	
	Max :	13691	1518	0													
	Min :	1368	0	0													

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 378.75 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 398.75 kN/m^2

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 393.36 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 413.36 kN/m^2

Sample calculation of values of base pressure:

Load case : D + F1 + W

Maximum pressure, p_1 (pmax) = 405.75+14.61+0.00= 420.36 kN/m^2

p_2= 405.75+14.61-0.00= 420.36 kN/m^2

p_3= 405.75-(14.61+0.00)= 391.14 kN/m^2

p_4= 405.75-14.61+0.00= 391.14 kN/m^2

Maxm pressure excluding weight of soil (p_{max}-γh)= 420.36-1.5*18= 393.36 kN/m^2

Maximum average Pressure along X, $p_{avx(max)}$ = (420.36+420.36)/2 = 420.36 kN/m^2

Minimum average Pressure along X, $p_{avx(minx)}$ = (391.14+391.14)/2 = 391.14 kN/m^2

Maximum average Pressure along Y, $p_{avy(max)}$ = (420.36+391.14)/2 = 405.75 kN/m^2

Minimum average Pressure along Y, $p_{avy(minx)}$ = (391.14+420.36)/2 = 405.75 kN/m^2

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DESIGN OF FOUNDATION MKD. F2

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((3+2*0.5*0.62)+(3+2*0.5*0.62))*2

= 14.48 m

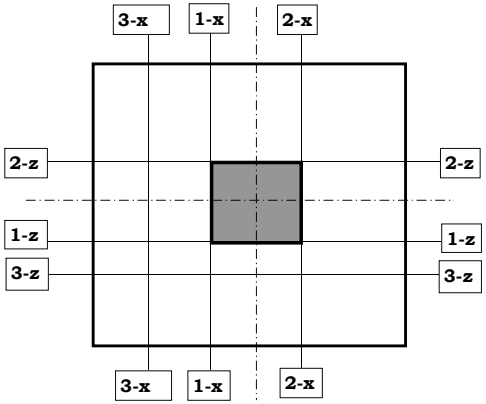
Punching Area= ((3+2*0.5*0.62)*(3+2*0.5*0.62))

= 13.1 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.7*24+0.8*18)

= 31.20 kN/m^2

Area of footing, A_f= 25.00 m^2



Load Case	Load Combination	p_{1x}	p_{2x}	p_{3x}	$B.M_{1x}$	$B.M_{2x}$	$S.F_{3x}$	$S.F_{(punching)}$
Ultimate Conditions								
301	1.4 (D + F1)	547.62	547.62	547.62	258.21	258.21	185.91	6145.45
302	1.2(D + F1) + 1.6 W	490.72	462.66	496.70	232.88	212.61	168.19	5301.31
303	1.2(D + F2) + E	520.40	432.97	539.06	254.32	191.17	184.72	5301.31
304	0.9 D + 1.6 W	70.57	42.52	76.55	22.80	2.54	16.93	301.59
305	0.9 (D+ F2) + E	358.60	271.16	377.25	173.41	110.27	126.47	3375.79

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DESIGN OF FOUNDATION MKD. F2

Sample calculation of BM & SF

Load case : 301

p_{1x} = 547.62-(547.62-547.62)*1/5 = 547.62 kN/m^2 Overhang = 1 m

p_{2x} = 547.62+(547.62-547.62)*1/5 = 547.62 kN/m^2 Effective depth = 0.64 m

p_{3x} = 547.62-(547.62-547.62)*(1-0.64)/5 = 547.62 kN/m^2

BM_{1x} = (547.62-31.2)*1*1/2+0.5*(547.62-547.62)*1*1*0.67 = 258.21 kN-m/m

BM_{2x} = (547.62-31.2)*1*1/2-0.5*(-547.62+547.62)*1*1*0.67 = 258.21 kN-m/m

SF_{3x} = (547.62-31.2)*(1-0.64)+0.5*(547.62-547.62)*(1-0.64) = 185.91 kN/m

$S.F_{(punching)}$ = (25-13.1)*(547.62-31.2) = 6145.45 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	258.21	0.63	0.84	0.002	1299.43	1152.00	20ϕT	241.77	300	225	1396.26
At top	0.00	0.00	0.00	0.000	0.00	384.00	12ϕT	294.52	300	250	452.39

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 185.91 kN/m

Shear stress , τ_v= (185.91*1000)/(1000*640)

= 0.29 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 6145.45 kN

Strength of Concrete in shear Vc1 = 27412.75 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 13176.29 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 18275.16 kN **Table 22.6.5.2_ (c)**

Design Vc= 13176.29 kN

Allowable shear stress, (ϕ Vc)/bd = 1.07 N/mm^2

Design punching shear stress, τ = (6145.45*1000)/(14480*640)

= 0.66 N/mm^2 **No shear reinforcement reqd.**

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DESIGN OF FOUNDATION MKD. F2

Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z}	p_{2z}	p_{3z}	$B.M_{1z}$	$B.M_{2z}$	$S.F_{3z}$
Ultimate Conditions							
301	1.4 (D + F1)	547.62	547.62	547.62	258.21	258.21	196.24
302	1.2(D + F1) + 1.6 W	476.69	476.69	476.69	222.75	222.75	169.29
303	1.2(D + F2) + E	476.69	476.69	476.69	222.75	222.75	169.29
304	0.9 D + 1.6 W	56.55	56.55	56.55	12.68	12.68	9.63
305	0.9 (D+ F2) + E	314.88	314.88	314.88	141.84	141.84	107.80

Sample calculation of BM & SF

Load case : 301

p_{1z} = 547.62-(547.62-547.62)*1/5= 547.62 kN/m^2

p_{2z} = 547.62+(547.62-547.62)*1/5= 547.62 kN/m^2 Overhang = 1 m

p_{3z} = 547.62-(547.62-547.62)*(1-0.62)/5= 547.62 kN/m^2 Effective depth = 0.62 m

BM_{1z} = (547.62-31.2)*1*1/2+0.5*(547.62-547.62)*1*1*0.67 = 258.21 kN-m/m

BM_{2z} = (547.62-31.2)*1*1/2-0.5*(-547.62+547.62)*1*1*0.67 = 258.21 kN-m/m

SF_{3z} = (547.62-31.2)*(1-0.62)+0.5*(547.62-547.62)*(1-0.62) = 196.24 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	258.21	0.672	0.896	0.0022	1385.95	1116.00	20ϕT	226.67	300	225	1396.3
At top	0.00	0.000	0.000	0.0000	0.00	372.00	12ϕT	304.03	300	250	452.4

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MD-216-3320-EG-CV-CAL-1210	Mansourah Massarah Gold Project	
900707_GAU01_ST02_CED01_ 33201_OU100665184_R3	DESIGN CALCULATION LOW GRADE THICKENING DESIGN OF FOUNDATION	Date: 15.11.2023
O19045_450_C_K_24088_001_R3	AS BUILT	Rev: D01

DESIGN OF FOUNDATION MKD. F2

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 196.24 kN/m

Shear stress , τ_v= (196.24*1000)/(1000*620)

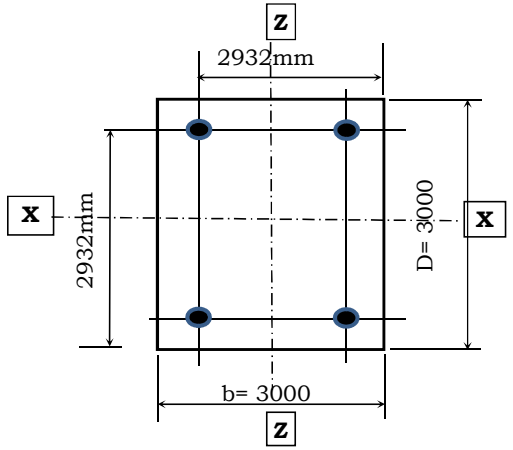
= 0.32 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :				
	At Bottom	20ϕT	225	c/c
	At Top	12ϕT	250	c/c
Along Z Direction :				
	At Bottom	20ϕT	225	c/c
	At Top	12ϕT	250	c/c

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MD-216-3320-EG-CV-CAL-1210	Mansourah Massarah Gold Project	
900707_GAU01_ST02_CED01_ 33201_OU100665184_R3	DESIGN CALCULATION LOW GRADE THICKENING DESIGN OF FOUNDATION	Date: 15.11.2023
O19045_450_C_K_24088_001_R3	AS BUILT	Rev: D01

DESIGN OF COLUMN MKD C5 **(Central Column)**



Input:

Clear cover to reinf.=	50mm
Dia. of main reinf.=	16mm
Dia. of link=	10mm
Effective cover =	68mm
Grade of Concrete (f_c') =	M35
Grade of Steel (f_y) =	Fy420
Grade of Reinf(links).=	Fy420
Min. Steel Ratio	0.15%
Max. Steel Ratio	4.00%

Table 20.6.1.3.1

For Compression-controlled sections

Strength reduction factor ϕ = 0.65 Table 21.2.2

Unsupproted length of column about X (l_{ux}) =	3000mm
Unsupproted length of column about Z (l_{uz}) =	3000mm

For Columns not braced against sidesway,

Where, Effective length factor K= 1.2

Slenderness ratio, $K\,l_{ux} / r_x$ = 4.16 < 22 SECTION 6.2.5a Radius of gyration r_x= 866 mm

$K\,l_{uz} / r_z$ = 4.16 < 22 SECTION 6.2.5a Radius of gyration r_z= 866 mm

MD-216-3320-EG-CV-CAL-1210	Mansourah Massarah Gold Project	
900707_GAU01_ST02_CED01_ 33201_OU100665184_R3	DESIGN CALCULATION LOW GRADE THICKENING DESIGN OF FOUNDATION	Date: 15.11.2023
O19045_450_C_K_24088_001_R3	AS BUILT	Rev: D01

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	Tank Dead Load (D)	152.0	0.0	0.0	0.0	0.0
2	Max. Fluid Load (F1)	8715.0	0.0	0.0	0.0	0.0
3	Normal Opl Fluid Load (F2)	7176.0	0.0	0.0	0.0	0.0
4	Wind Load (W)(+/-)	0.0	54.0	0.0	110.0	0.0
5	Earthquake Load (E)(+/-)	0.0	270.0	0.0	546.0	0.0
			γ_x =	0.96		
			γ_z =	0.96		

Load Case	Load Combination	Factored Load/Moment			Nominal Load/Moment			Kn = Pn/f'cbD (N/mm^2)
		Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
301	1.4 (D + F1)	1241.38	0.0	0.0	1909.8	0.0	0.0	0.06
302	1.2(D + F1) + 1.6 W	1064.0	43.5	0.0	1637.0	67.0	0.0	0.05
303	1.2(D + F2) + E	1064.0	135.6	0.0	1637.0	208.6	0.0	0.05
304	0.9 D + 1.6 W	13.7	43.5	0.0	21.0	67.0	0.0	0.00
305	0.9 (D+ F2) + E	659.5	135.6	0.0	1014.6	208.6	0.0	0.03

Reinforcement: Provide,

4-Nos	16 mm Dia
36-Nos	16 mm Dia
36-Nos	16 mm Dia

804.57 Sq mm (At Corner)
7241.14 Sq mm (Along Z axis)
7241.14 Sq mm (Along Y axis)

A_{sc} pvd. = 15286.86 Sq mm

Total A_{sc} = 0.17 %

β = P_n / P_0 (range is 0.55 to 0.70) Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 27371.57 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.9		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f'_cb^2D$	$R_{nx}=M_{nox}/f'_cb^2D$						
301	0.06	0.06	5670.0	5670.0	0.55	1.16	0.00	O.K
302	0.06	0.06	5670.0	5670.0	0.55	1.16	0.01	O.K
303	0.06	0.06	5670.0	5670.0	0.55	1.16	0.02	O.K
304	0.03	0.03	2835.0	2835.0	0.55	1.16	0.01	O.K
305	0.04	0.04	3780.0	3780.0	0.55	1.16	0.03	O.K

Links, 12mm dia @ 150 C/C

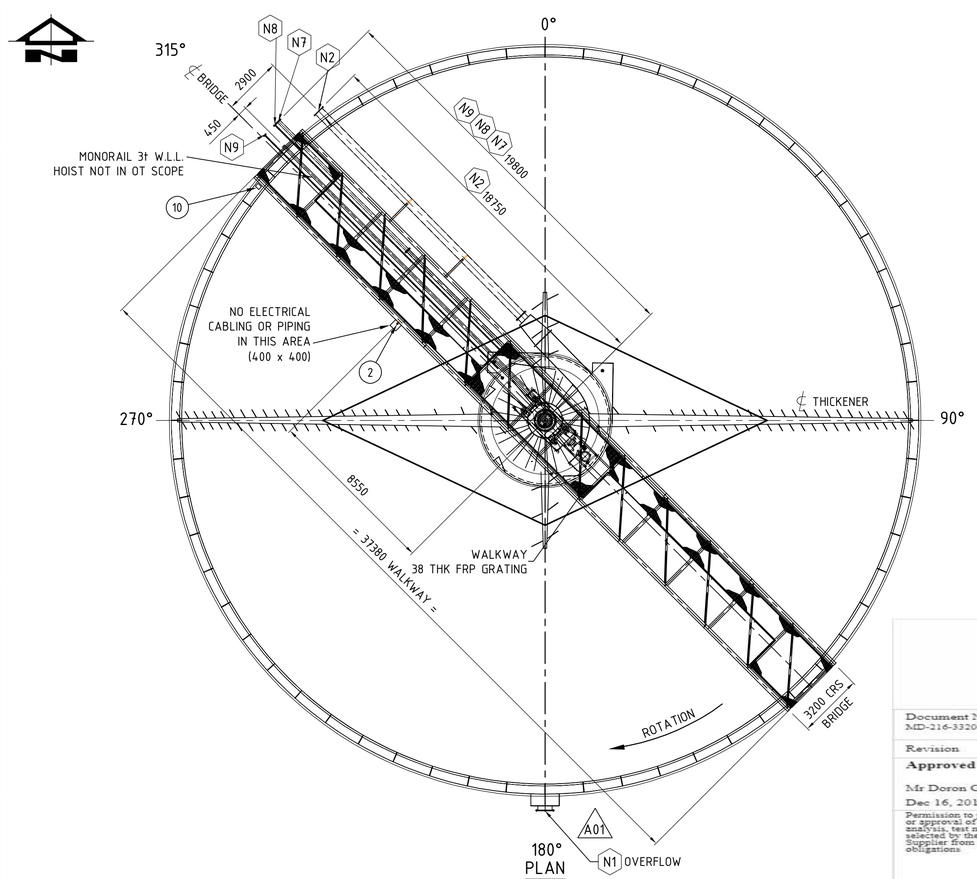
REFERENCE

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

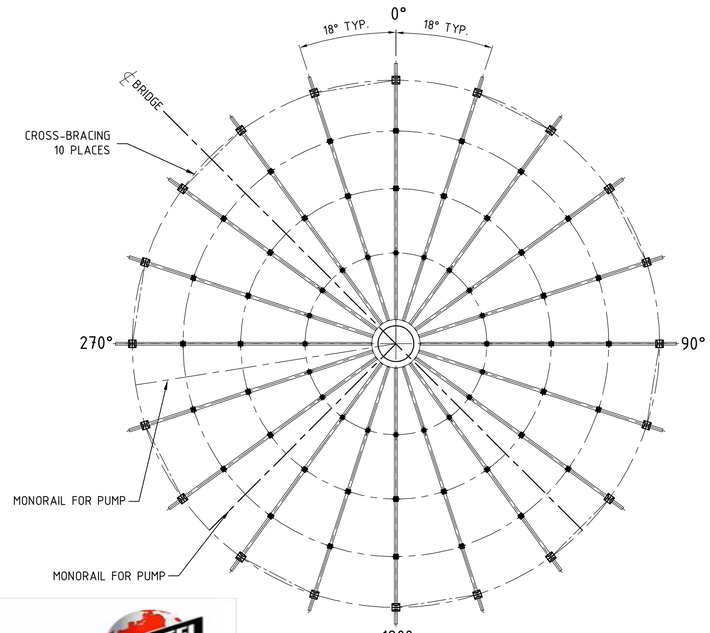
ATTACHMENT 1 LOAD DATA DRAWINGS

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DRAWING NUMBER MD - 216 - 3320 - EG- ME - GAD - 1001 REV A02



180° PLAN



PLAN VIEW
SHOWING RADIAL BEAMS ARRANGEMENT

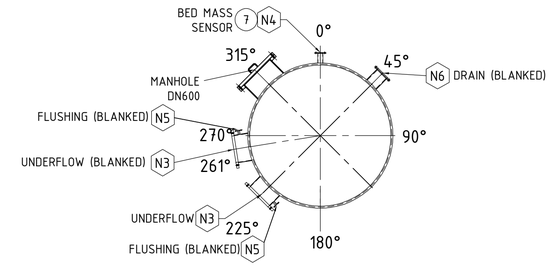
SCALE 1:250

ITEM	DESCRIPTION	SIZE	Q.TY	CONNECTION TYPE
N9	MUD DIVER SPRAY WATER	DN50	1	2"ASME B16.5-150 lbs
N8	FLOC CONCENTRATE HEADER	DN40	1	1-1/2"ASME B16.5-150 lbs
N7	FLOC DILUTION HEADER	DN100	1	4"ASME B16.5-150 lbs
N6	DRAIN (BLANKED)	DN400	1	16"ASME B16.5-150 lbs
N5	FLUSH (BLANKED)	DN80	2	3"ASME B16.5-150 lbs
N4	BED MASS	DN80	1	3"ASME B16.5-150 lbs
N3	UNDERFLOW (1 DUTY /1 STANDBY)	DN450	2	18"ASME B16.5-150 lbs
N2	FEED PIPE	DN400	1	16"ASME B16.5-150 lbs
N1	OVERFLOW	DN700	1	28"ASME B16.47-150 lbs

NOZZLE LIST

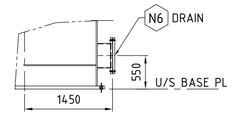
PART	DESCRIPTION	SUB-DWG.NO.\ MATERIAL	QTY	WEIGHT [KG]	ITEM NO.	REMARKS
10	TURBIDITY SENSOR	-	1	-	-	
9	POWER PACK - 11 kW	-	1	-	-	
8	CONTROL PANEL	-	1	-	-	
7	BED MASS SENSOR	-	1	-	-	
6	FEED ASSEMBLY	MD-216-3320-EG-ME-DED-1006	1	-	-	
5	RAKE MECHANISM ASSEMBLY	MD-216-3320-EG-ME-DED-1005	1	11944	-	
4	BRIDGE ASSEMBLY	MD-216-3320-EG-ME-DED-1004	1	-	-	
3	TANK & SUPPORT STRUCTURE	MD-216-3320-EG-ME-DED-1001	1	139768	-	
2	BED LEVEL SENSOR	-	1	-	-	
1	MEGA DRIVE ASSEMBLY - (300 LIFT - BONFIGLIOLI 319 L4 FZ SAI SERIES)	MD-216-3320-EG-ME-DED-1007	1	-	-	

NOTE: ACTUAL WEIGHTS TO BE CONFIRMED



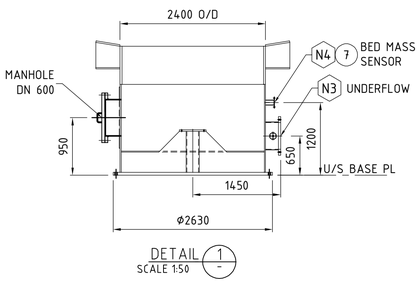
UNDERFLOW BOOT DETAILS

SCALE 1:50

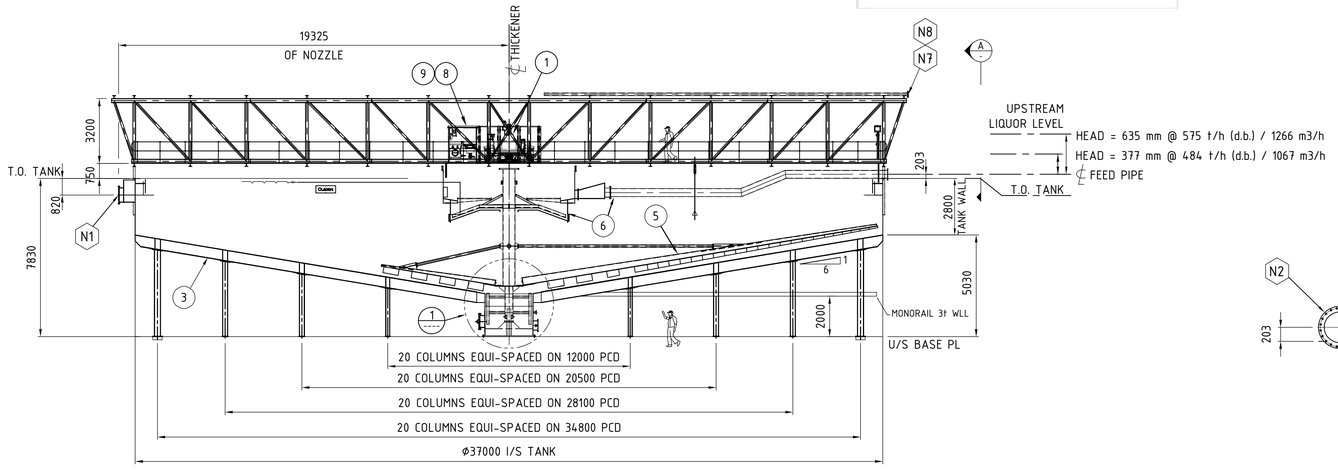


DRAIN NOZZLE DETAILS

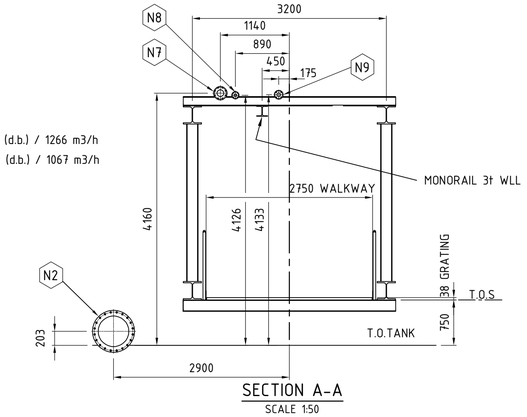
SCALE 1:50



DETAIL 1
SCALE 1:50



SECTIONAL ELEVATION
REFER TO PLAN FOR TRUE ORIENTATION



SECTION A-A
SCALE 1:50

SPECIAL MATERIALS AND COATINGS	
UNDERFLOW CONE	CARBON STEEL PAINTED
UNDERFLOW NOZZLES	CARBON STEEL PAINTED
TANK SHELL	CARBON STEEL PAINTED
TANK SUPPORTS	CARBON STEEL PAINTED
MECHANISM	CARBON STEEL PAINTED
FEEDPIPE	CARBON STEEL PAINTED, INTERNAL RUBBER LINING
FEEDWELL	CARBON STEEL PAINTED, PARTIAL INTERNAL RUBBER LINING
FLOC HEADER	316L
WALKWAY GRATING	FRP
FROTH BAFFLE	304SS

Document Number
MD-216-3320-EG-ME-GAD-1001
Revision A02
Approved
Mr Doron Golan - Bechtel Global
Dec 16, 2019, 5:34 PM GMT+2:00
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DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3320-EG-ME-DED-1001	GENERAL NOTES								
MD-216-3320-EG-ME-SPC-3001	TECHNICAL SPECIFICATION								
MD-216-3320-EG-ME-DAT-1001	EQUIPMENT DATASHEET	A02	22.11.2019	REVISED AS PER CLIENT COMMENTS	TK	AC	SC	RL	
MD-216-3320-EG-ME-GAD-1002	FOOTING AND LOADING PLAN	A01	04.10.2019	ISSUED FOR APPROVAL	TK	AC	SC	RL	

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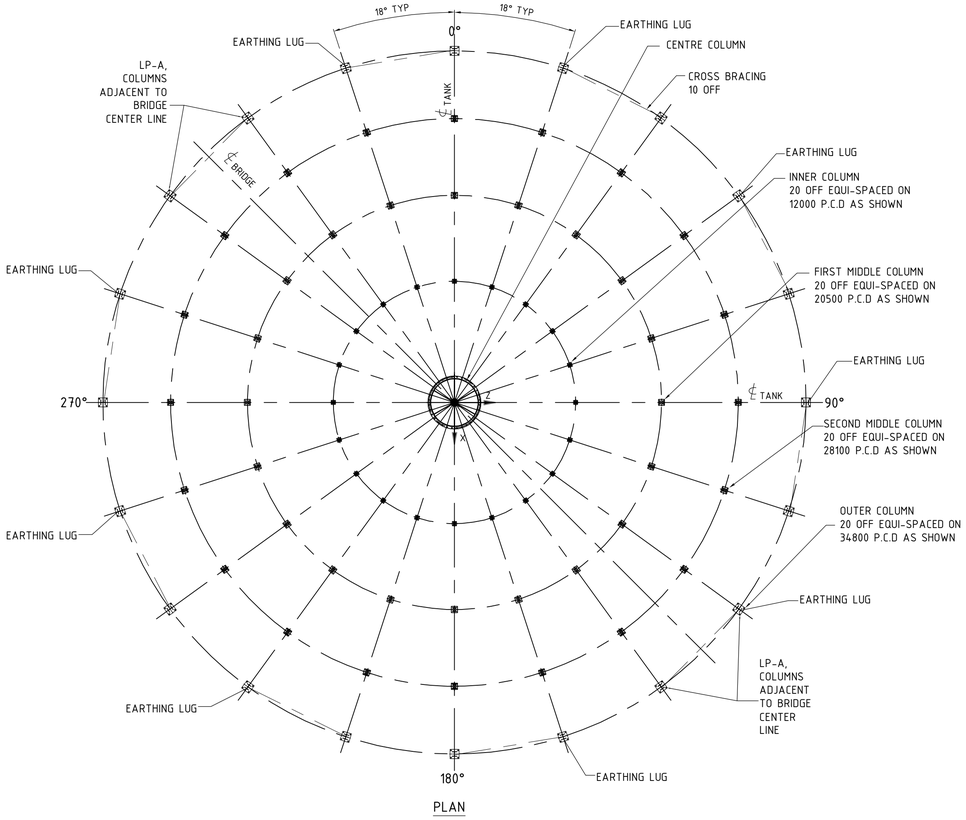
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OUTOTEC DRAWING NUMBER: 900707__GAU01__ST02__MTL03__33201 — DOCUMENT ID: OU601263184 — REVISION: 2

DRAWN	timkar	20.11.2019
CHECKED	alecha	20.11.2019
ENGINEER	soucha	22.11.2019
APPROVED	ricleh	22.11.2019
EMPLOYER		
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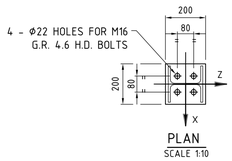
MANSOURAH - MASSARAH GOLD PROJECT
EQUIPMENT GENERAL ARRANGEMENT DRAWING
LOW GRADE THICKENING
3320-THK-001 T20 - LOW GRADE THICKENER

DRAWING NUMBER: MD - 216 - 3320 - EG- ME - GAD - 1001 — REV: A02

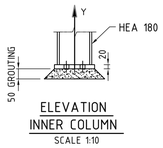
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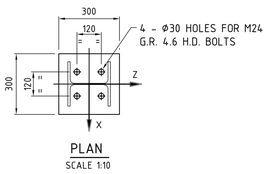
PLAN



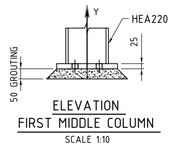
PLAN
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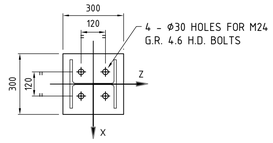
ELEVATION
INNER COLUMN
SCALE 1:10



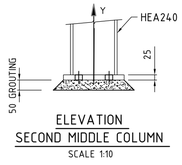
PLAN
SCALE 1:10



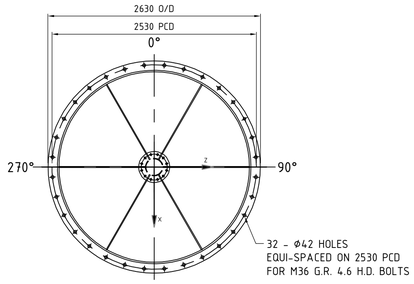
ELEVATION
FIRST MIDDLE COLUMN
SCALE 1:10



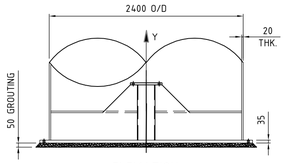
PLAN
SCALE 1:10



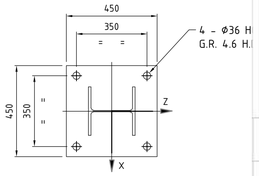
ELEVATION
SECOND MIDDLE COLUMN
SCALE 1:10



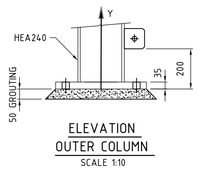
PLAN
SCALE 1:25



ELEVATION
CENTRE COLUMN
SCALE 1:25



PLAN
SCALE 1:10



ELEVATION
OUTER COLUMN
SCALE 1:10

LOAD TABLE

	Column PCD (mm)	Tank Dead Load	Max Fluid Load SG 1.648		Normal Operating Fluid Load SG 1.357		Seismic Load (+/-)			Wind Load (+/-)			Bridge/Drive Dead Load (LP-A) *NOTE 9	Bridge/Drive Live Load (LP-A) *NOTE 9
		Vertical (Y) (kN)	Horizontal (X,Z) (kN)	Vertical (Y) (kN)	Horizontal (X,Z) (kN)	Vertical (Y) (kN)	Horizontal (X,Z) (kN)	Vertical (Y) (kN)	Moment (X,Z) (kNm)	Horizontal (X,Z) (kN)	Vertical (Y) (kN)	Moment (X,Z) (kNm)	Vertical (Y) (kN)	Vertical (Y) (kN)
Outer Columns	34 800	34	112	680	93	560	26	44	0	9	16	0	123	167
2nd Middle Columns	28 100	6	0	822	0	677	0	3	0	0	2	0	0	0
1st Middle Columns	20 500	10	0	784	0	646	0	0	0	0	1	0	0	0
Inner Columns	12 000	8	0	582	0	479	0	0	0	0	1	0	0	0
Central Column	0	152	0	8 715	0	7 176	270	0	546	54	0	110	0	0
Total Applied Load	-	1 316	0	66 087	0	54 418	561	0	2 639	112	0	625	397	541

NOTES:

1. ALL LOADS LISTED ARE PRIMARY UNFACTORED LOADS SUITABLE FOR LOAD COMBINATIONS TO SBC301
2. SEISMIC LOADS ARE CALCULATED CONSIDERING DYNAMIC SLOSHING EFFECTS OF NORMAL OPERATING SG LIQUIDS
3. COMBINATIONS INCLUDING SEISMIC LOAD TO USE NORMAL OPERATING FLUID LOAD
4. ALL HORIZONTAL REACTIONS ARE IN THE PLANE OF BRACING
5. TOTAL LOADS AND MOMENTS ARE CALCULATED AT THE UNDERSIDE OF BASEPLATE
6. COLUMN LOADS ARE GIVEN PER COLUMN
7. DIFFERENTIAL SETTLEMENT BETWEEN ANY ADJACENT COLUMN SHALL BE LIMITED TO L / 600
8. TILTING OF THE TANK DUE TO OUTER COLUMN DIFFERENTIAL SETTLEMENT SHALL BE LIMITED TO 14 mm
9. BRIDGE/DRIVE LOADS REACTED AT LOAD POINTS A (LP-A) ONLY. LP-A IS ADJACENT TO BRIDGE CENTER LINE.
10. LENGTH OF ANCHOR BOLTS TO BE DEFINED AS A FUNCTION OF CIVIL ENGINEERING.

SEISMIC DATA TO SBC301

Ss, S1:	0.10, 0.02
Importance Factor:	1
Site Class:	B

WIND DATA TO SBC301

V:	43	m/s
Exposure Class:	C	

FOUNDATION TOLERANCES

HEIGHT OF COLUMN FOUNDATIONS	± 10 mm
ANCHOR BOLTS - DISTANCE BETWEEN ANY BOLT IN A GROUP	± 3 mm
ANCHOR BOLTS - DISTANCE BETWEEN ANY BOLT GROUP	± 6 mm
ANCHOR BOLTS - MAX TOLERANCE ACCUMULATION BETWEEN BOLT GROUPS	± 6 mm

BECHTEL

Document Number
MD-216-3320-EG-ME-GAD-1002

Revision A01

Approved with Comments

Mr Doron Golan - Bechtel Global

Dec 16, 2019, 3:54 PM GMT+2:00

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DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3000-EG-ME-DED-1001	GENERAL NOTES								
MD-216-3000-EG-ME-SPC-3001	TECHNICAL SPECIFICATION								
MD-216-3320-EG-ME-DAT-1001	EQUIPMENT DATASHEET								
MD-216-3320-EG-ME-GAD-1001	EQUIPMENT GENERAL ARRANGEMENT DRAWING	A01	14.11.2019	ISSUED FOR APPROVAL	TK	AC	SC	RL	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO DRAWING NUMBER: 019045-450-M-D-02472-001 REVISION: 0

OUTOTEC DRAWING NUMBER: 900707 __GAU01 __ST02__MLD15 __33201 DOCUMENT ID: OU601242858 REVISION: 1

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DRAWN	timkar	13.11.2019	
CHECKED	alecha	13.11.2019	
ENGINEER	soucha	14.11.2019	
APPROVED	ricleh	14.11.2019	
EMPLOYER			
SCALE	1:150		A1

MANSOURAH - MASSARAH GOLD PROJECT
FOUNDATION AND LOAD PLAN
37m LOW GRADE THICKENER
3320-THK-001 T20-LOW GRADE THICKENING

DRAWING NUMBER: MD - 216 - 3320 - EG - ME - GAD - 1002 REV: A01

ATTACHMENT 2 STAAD OUTPUT

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**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6          *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    MAR 24, 2020                *
*           Time=    15:36: 5                    *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: D:\DESIGN\Ma'aden_MMGP\Process Area Design\Flotation Tailing Thikening 3320\A02\Staad\F1.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 21-JAN-20
     4. JOB NAME MANSOURAH MASSARAH GOLD PROJECT
     5. JOB CLIENT MA'ADEN
     6. JOB NO 3320
     7. ENGINEER NAME AVG
     8. CHECKER NAME SJM
     9. APPROVED NAME AG
    10. END JOB INFORMATION
    11. INPUT WIDTH 79
    12. UNIT METER KN
    13. JOINT COORDINATES
    14. 1 0 0 0; 2 0.85 0 0; 3 4.2 0 0; 4 8 0 0; 5 12.25 0 0; 6 13 0 0
    15. MEMBER INCIDENCES
    16. 1 1 2; 2 2 3; 3 3 4; 4 4 5; 5 5 6
    17. DEFINE MATERIAL START
    18. ISOTROPIC CONCRETE
    19. E 2.17185E+007
    20. POISSON 0.17
    21. DENSITY 23.5616
    22. ALPHA 1E-005
    23. DAMP 0.05
    24. TYPE CONCRETE
    25. STRENGTH FCU 27579
    26. END DEFINE MATERIAL
    27. MEMBER PROPERTY AMERICAN
    28. 1 TO 5 PRIS YD 0.7 ZD 1.5
    29. CONSTANTS
    30. MATERIAL CONCRETE ALL
    31. SUPPORTS
    32. 2 TO 5 FIXED BUT MY MZ KFY 0.2
    33. LOAD 301 LOADTYPE NONE  TITLE 1.4 ( D + F1)
    34. JOINT LOAD
    35. 2 FY -1171.8 MZ 564.48
    36. 3 FY -1159.2
    37. 4 FY -1111.6
    38. 5 FY -826
```

```
39. MEMBER LOAD
40. 1 TO 5 UNI GY -47.85
41. 1 TRAP GY 548.96 526.247
42. 2 TRAP GY 526.247 436.733
43. 3 TRAP GY 436.733 335.194
44. 4 TRAP GY 335.194 221.631
45. 5 TRAP GY 221.631 201.59
46. LOAD 302 LOADTYPE NONE  TITLE 1.2( D + F1 ) + 1.6 W
47. JOINT LOAD
48. 2 FY -1020.4 MZ 535.68
49. 3 FY -995.6
50. 4 FY -953.8
51. 5 FY -709
52. MEMBER LOAD
53. 1 TO 5 UNI GY -47.85
54. LOAD 303 LOADTYPE NONE  TITLE 1.2( D + F2 ) + E
55. JOINT LOAD
56. 2 FY -904.4 MZ 495.36
57. 3 FY -822.6
58. 4 FY -787.2
59. 5 FY -584.4
60. MEMBER LOAD
61. 1 TO 5 UNI GY -47.85
62. 1 TRAP GY 437.69 417.758
63. 2 TRAP GY 417.758 339.203
64. 3 TRAP GY 339.203 250.096
65. 4 TRAP GY 250.096 150.437
66. 5 TRAP GY 150.437 132.85
67. LOAD 304 LOADTYPE NONE  TITLE 0.9 D + 1.6 W
68. JOINT LOAD
69. 2 FY -157.3 MZ 51.84
70. 3 FY -7.4
71. 4 FY -10
72. 5 FY -8.2
73. MEMBER LOAD
74. 1 TO 5 UNI GY -47.85
75. LOAD 305 LOADTYPE NONE  TITLE 0.9 ( D+ F2 ) + E
76. JOINT LOAD
77. 2 FY -689.3 MZ 394.92
78. 3 FY -617.7
79. 4 FY -590.4
80. 5 FY -438.3
81. MEMBER LOAD
82. 1 TO 5 UNI GY -47.85
83. 1 TRAP GY 348.1 332.21
84. 2 TRAP GY 332.21 269.583
85. 3 TRAP GY 269.583 198.543
86. 4 TRAP GY 198.543 119.091
87. 5 TRAP GY 119.091 105.07
88. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 6 NUMBER OF MEMBERS 5
NUMBER OF PLATES 0 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 4

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 1/ 1/ 9 DOF
TOTAL PRIMARY LOAD CASES = 5, TOTAL DEGREES OF FREEDOM = 24
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 50486.0 MB

89. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAR 24,2020 TIME= 15:36: 8 ****

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*   For technical assistance on STAAD.Pro, please visit    *
*   http://selectservices.bentley.com/en-US/               *
*                                                          *
*   Details about additional assistance from               *
*   Bentley and Partners can be found at program menu      *
*   Help->Technical Support                                *
*                                                          *
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*                http://www.bentley.com                    *
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